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AMERICAN SOCIETY OF CIVIL
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TRANSACTIONS

Paper No. 1192

**EXPERIMENTS ON RETAINING WALLS AND
PRESSURES ON TUNNELS.**

BY WILLIAM CAIN, M. AM. SOC. C. E.

WITH DISCUSSION BY MESSRS. J. R. WORCESTER, J. C. MEEM, AND WILLIAM
CAIN.

The most extended experiments relating to retaining walls are those pertaining to retaining walls proper and the more elaborate ones on small rotating retaining boards. The results referring to the former agree fairly well with a rational theory, especially when the walls are several feet in height; but with the latter, many discrepancies occur, for which, hitherto, no explanation has been offered.

It will be the main object of this paper to show that the results of these experiments on small retaining boards can be harmonized with theory by

The result of the investigation will then be to give to the practical constructor more confidence in the theory of the sliding prism, which serves as the basis of the methods to follow.

produce the inner face of the wall to meet the surface of the surcharge at F . It is desired to find the thrust against the plane, AF , for 1 lin. ft. of the wall. Draw AD through A , the foot of the wall, making the angle of repose, ϕ , of the earth with the horizontal and meeting the upper surface at D . Since any possible prism of rupture, as $FACR$, in tending to move downward, develops friction against both surfaces, AC and AF , the earth thrust on the wall will make an angle, ϕ' , with the normal to AF , where ϕ' is the angle of friction of the earth on the wall. As the earth settles more than the wall, this friction will always be exerted. Again, as the wall, from its elasticity and that of the foundation, will tend to move over at the top on account of the earth thrust, the earth, with its frictional grip on the wall, will tend to



prevent this, so that the friction is exerted downward in either case, and the direction of the earth thrust, E , on AF is as given in Fig. 1.

However, if $\phi' > \phi$, a thin slice of earth will move with the wall, and the rubbing will be that of earth on earth, so that ϕ' in this case must be replaced by ϕ . This rule will apply in all cases that follow, without further remark, wherever ϕ' is mentioned.

Now draw Ay , making the angle, $\phi + \phi'$, with AF , as shown; then draw FB parallel to AR , to the intersection, B , with DR produced. From B a parallel to Ay is constructed, meeting AD at O .

Since theory gives the relation: $AI = \sqrt{AD \cdot AO}$, two constructions follow, by geometry, for locating the point, I . By the first, a semicircle is described on AD as a diameter; at the point, O , a perpendicular is erected to AD , meeting the semicircle in M ; then AI is laid off equal to the chord, AM . By the second construction, a semicircle is described on OD as a diameter, a tangent to it, Az , from A is drawn, limited by the perpendicular radius, and finally AI is laid off equal to Az .

The point, I , having been thus found by either construction, draw IC parallel to Ay to the intersection, C , with RD . AC is the plane of rupture. On laying off $IL = IC$, and dropping the perpendicular, CH , from C on AD , the earth pressure, E , on AF is given by $\frac{1}{2}CI \cdot CH \cdot e$, where e is the weight of a cubic unit of the earth; otherwise, the value of E is given by e times the area of the shaded triangle, ICL . If the dimensions are in feet, and e is in pounds per cubic foot, the thrust, E , will be given in pounds.

In Figs. 2 and 3, the retaining boards, AB , are vertical, and BO is drawn, making the angle, $\phi + \phi'$, with the vertical, AB . The upper surface of the earth is BD , and the constructions for locating I and C are the same as for Fig. 1. AC , in all the figures, represents the plane of rupture.¹ In all cases, the earth thrust found as above is supposed to make the angle, ϕ' (as shown), with the normal to the inner wall surface.

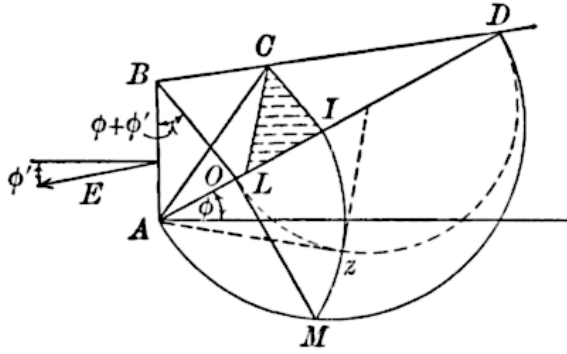


FIG. 2.

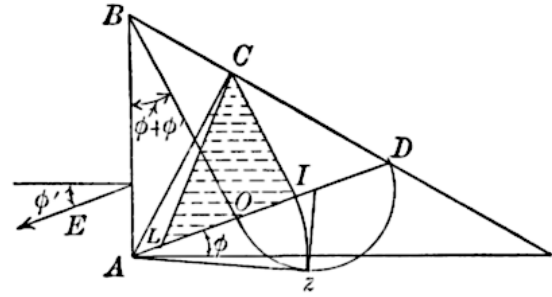


FIG. 3.

In the Rankine theory, pertaining, say, to [Fig. 2](#), the earth thrust on a vertical plane, AB , is always taken as acting parallel to the top slope. This is true for the pressure on a vertical plane in the interior of a mass of earth of indefinite extent, but it is not true generally for the pressure against a retaining wall. Thus, when BD , [Fig. 2](#), is horizontal, Rankine's thrust on AB would be taken as horizontal, which entirely ignores the friction of the earth on the wall. The two theories agree when $\phi' = \phi$ and BD slopes at the angle of repose, in which case, as BC is parallel to AI , there is no intersection, D . It is a limiting case in which, to compute the thrust, CIL can be laid off from any point in BD , on drawing CI parallel to BO , etc. As BD approaches the natural slope, the point, D , recedes indefinitely to the right, and it is seen that the plane of rupture, AC , approaches indefinitely the line, AD , or the natural slope. This limiting case, on account of the excessive thrust corresponding, will be examined more carefully in the sequel.

If h = the height of the wall, AB , in feet, and e = the weight of a cubic foot of earth, in pounds, then when $\phi' = \phi$, and the surface BD , [Fig. 2](#), slopes at the angle of repose, the earth thrust, in pounds, is given by the equation:

$$E = \frac{1}{2} e h^2 \cos \phi \quad (1)$$

If, however, ϕ' is not equal to ϕ , then E is directed at the angle, ϕ' , to the normal to the wall, and the thrust is:

$$E = \frac{1}{2} e h^2 \frac{\cos^2 \phi}{\cos \phi'} \quad (2)$$

The foregoing constructions, and the corresponding equations, are all derived from the theory of the sliding prism. The wedge, BAC , Figs. 2 and 3, is treated as an invariable solid, tending to slide down the two faces, AB and AC , at once, thus developing the full friction that can be exerted on these faces. In the case of actual rotation of the board, AB , it is found by experiment that each particle of earth in the prism, BAC , moves parallel to AC , each layer parallel to AC moving over the layer just beneath it.

A similar motion is observed if the board, AB , is moved horizontally to the left. However, in the first case (of rotation) the particles at A do not move at all, whereas in the second (of sliding motion) the particles about A move, rubbing over the floor, which thus resists the motion by friction. A thrust, thus recorded by springs or other device, in the case where the wall moves horizontally, would give an undervaluation at the lower part of AB and consequently the computed center of pressure on AB would be too high. On that account, only the experiments on rotating boards will be considered in this paper.

The theory of the sliding wedge, however, is justified, because no motion of either kind is actually supposed. The wedge, BAC , is supposed to be just on the point of motion, it being in equilibrium under the action of its weight, the normal components of the reactions of the wall, and the plane, AC , and all the friction that can be exerted along AB and AC . These forces remain the same, whatever incipient motion is supposed. The hypothesis of a plane surface of rupture, however, is not exactly realized, experiment showing that the earth breaks along a slightly curved surface convex to the moving mass. For the sake of simplicity, the theory neglects the cohesion acting, not only along AC , but possibly to a small extent along

AB. This additional force will be included in certain investigations to be given later.

These preliminary observations having been disposed of, the results of certain experiments on retaining walls at the limit of stability will now be given.

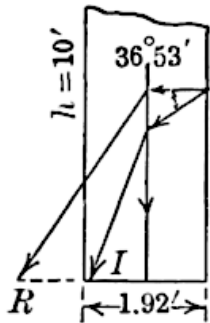


FIG. 4.

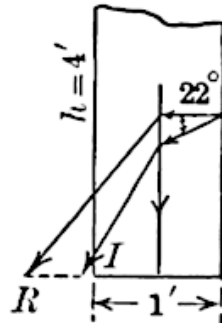


FIG. 5.

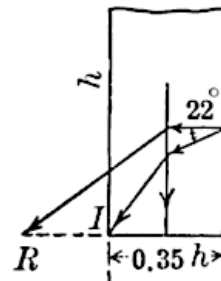


FIG. 6.

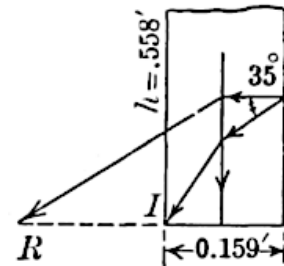


FIG. 7.

Figs. 4, 5, 6, and 7 refer to vertical rectangular walls backed by sand, except in the case of Fig. 5, where the filling was macadam screenings. The surface of the filling was horizontal in each case. To give briefly in detail the quantities pertaining to each wall, the following symbols will be used:

h = height of wall, in feet;

t = thickness of wall, in feet;

e = weight of filling, per cubic foot;

w = weight of wall, per cubic foot;

ϕ = angle of repose of filling;

ϕ' = angle of friction of filling on wall;

q = distance from center of pressure on the plane of the base to the outer toe, divided by t , using the theory that rationally includes the whole of the wall friction;

q_0 = the same, according to the Rankine theory.

q and q_0 are positive when the resultant on the base strikes within the base, otherwise they are negative.

Fig. 4 represents Lieut. Hope's wall of bricks laid in wet sand: $h = 10$, $t = 1.92$, $e = 95.5$, $w = 100$, $\phi = \phi' = 36^\circ 53'$. It was 20 ft. long, and was backed by earth level with its top. $q = +0.04$, $q_0 = -0.58$. The overhang, at the moment of failure, was probably 4 in. Including this, $q = -0.02$.

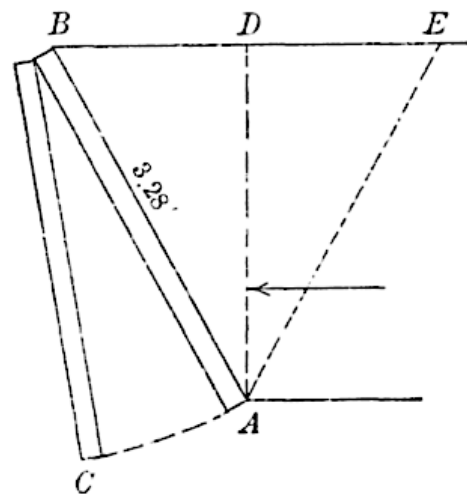
Fig. 5 shows Baker's wall of pitch-pine blocks, backed by macadam screenings, the level surface of which was 0.25 ft. below the top of the wall; $h = 4$, $t = 1$, $e = 101$, $w = 46$, $\phi = 39^\circ 48'$, $\phi' = 22^\circ$, the assumed angle of friction of timber on stone, $q = -0.06$, $q_0 = -0.55$.

Trautwine's experimental wall is shown in Fig. 6. Only the ratio of base to height, 0.35, was given by the author, but J. C. Trautwine, Jr., Assoc. Am. Soc. C. E., assures the writer that the walls were probably 6 in. in height, though certain notes refer to walls varying from about 4 to 9 in. $e = 89$, $w = 28$, $\phi = 33^\circ 42'$, ϕ' (assumed) $= 22^\circ$, $q = -0.03$, $q_0 = -0.75$.

The wall of Curie, Fig. 7, was of wood coated on the back by sand, so that $\phi = \phi' = 35^\circ$. Also, $h = 0.558$ ft., $t = 0.159$ ft., $\frac{e}{w} = 3.29$, $q = 0.02$, $q_0 = -1.32$.

These walls were all at the limit of stability, and the first two are of appreciable height, 10 ft. and 4 ft., respectively.

The figures show that the theory, including the whole of the wall friction, agrees fairly well with experiment, but that the Rankine theory does not thus agree. In both theories, the thrust, E , is supposed to act at one-third of the height from the base



of the wall to the surface of the filling; but,

FIG. 8.

in the Rankine theory, this thrust is

assumed to act horizontally, whereas, in the other theory, it is supposed to act in a direction making the angle, ϕ' , below the normal to the wall.

On combining the thrusts with the weight of the wall, as usual, the resultant strikes the base produced, at R in the first case (Rankine theory), but at I in the second case. Figs. 4 to 7 present a striking object lesson as to the inaccuracy of the Rankine method of treating experimental retaining walls.

In the next experiments, however, referring to a retaining structure consisting of two boards, hinged at the top, Fig. 8, and backed by sand level at the top, the Rankine theory is applicable when the board, AB , is placed either at or below the plane of rupture, on the left of AD . The thrust on AD is then assumed to act horizontally, at $\frac{1}{3}AD$ above A , and is combined with the weight of the sand, DAB , to find the resultant on the board. If the board is at the plane of rupture, this resultant will make the angle ϕ below the normal to AB ; hence, if one assumes a less thrust on AD , especially if inclined downward, the new resultant on AB will make an angle greater than ϕ with the normal to AB , which is inconsistent with stability.² The same reasoning applies when AB lies below the plane of rupture.³

The retaining board, 1 m. square, was coated with sand, so that $\phi = \phi' = 35^\circ$ for damp sand. Hence, for a horizontal thrust on AD , the plane of rupture (which bisects the angle between the vertical and the natural slope) makes an angle of $27^\circ 30'$ with the vertical. The board, AB , was set at this angle to the vertical, sand was filled in level with the top, and it was found that the structure was at the limit of stability when $AC = 0.45$ m. In the meantime, however, the sand had dried out, so that ϕ was $33^\circ 30'$; hence, strictly, the construction of Fig. 1 (for earth level with top of wall) applies; but, as the results can only differ inappreciably, the thrust on AD , acting horizontally, was computed for $\phi = 33^\circ 30'$ and

combined with the weight of sand, BAD , and the weight of structure, both acting through their centers of gravity, to find the resultant on the base, AC . It was found to cut it 0.11 of its width from the outer toe, C ; therefore $q = 0.11$.

In the next experiment, the angle, BAD , was 55° , $AC = 0.548$ m. and $\phi = 33^\circ 30'$. Pursuing the same method, it is found that $q = 0.02$, or the resultant on the base passes practically through C . The third experiment was on a smaller retaining board. Here $AB = 0.2$ m., $BAD = 55^\circ$, $\phi = \phi' = 35^\circ$, and $q = 0.06$.

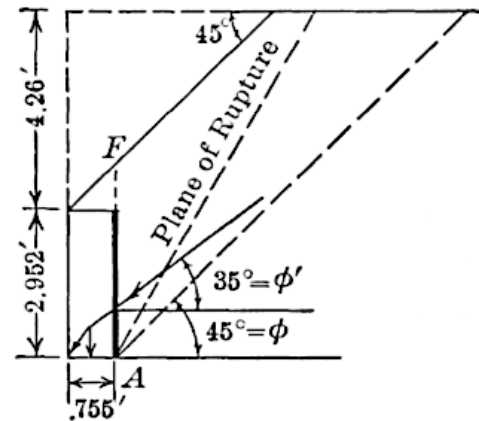


FIG. 9.

In Fig. 9 is shown a surcharged wall of Curie's, just at the limit of stability, having $h = 2.952$ ft., $t = 0.755$ ft. and the level upper surface of the surcharge being 4.26 ft. above the top of the wall. The surcharge extended over the wall at the angle, $\phi = 45^\circ$, corresponding to damp sand. Experiment gave $\phi' = 35^\circ$. The wall was of brick in Portland cement. The ratio, $\frac{e}{w} = 0.62$. It was found, using the construction of Fig. 1, that taking the thrust, E , as acting 1.24 ft. above the base, or at one-third of the height of the surface, AF , that $q = +0.03$; and further, that if E acts 1.303 ft. above the base, the resultant on the base passes exactly through the outer toe of the wall.

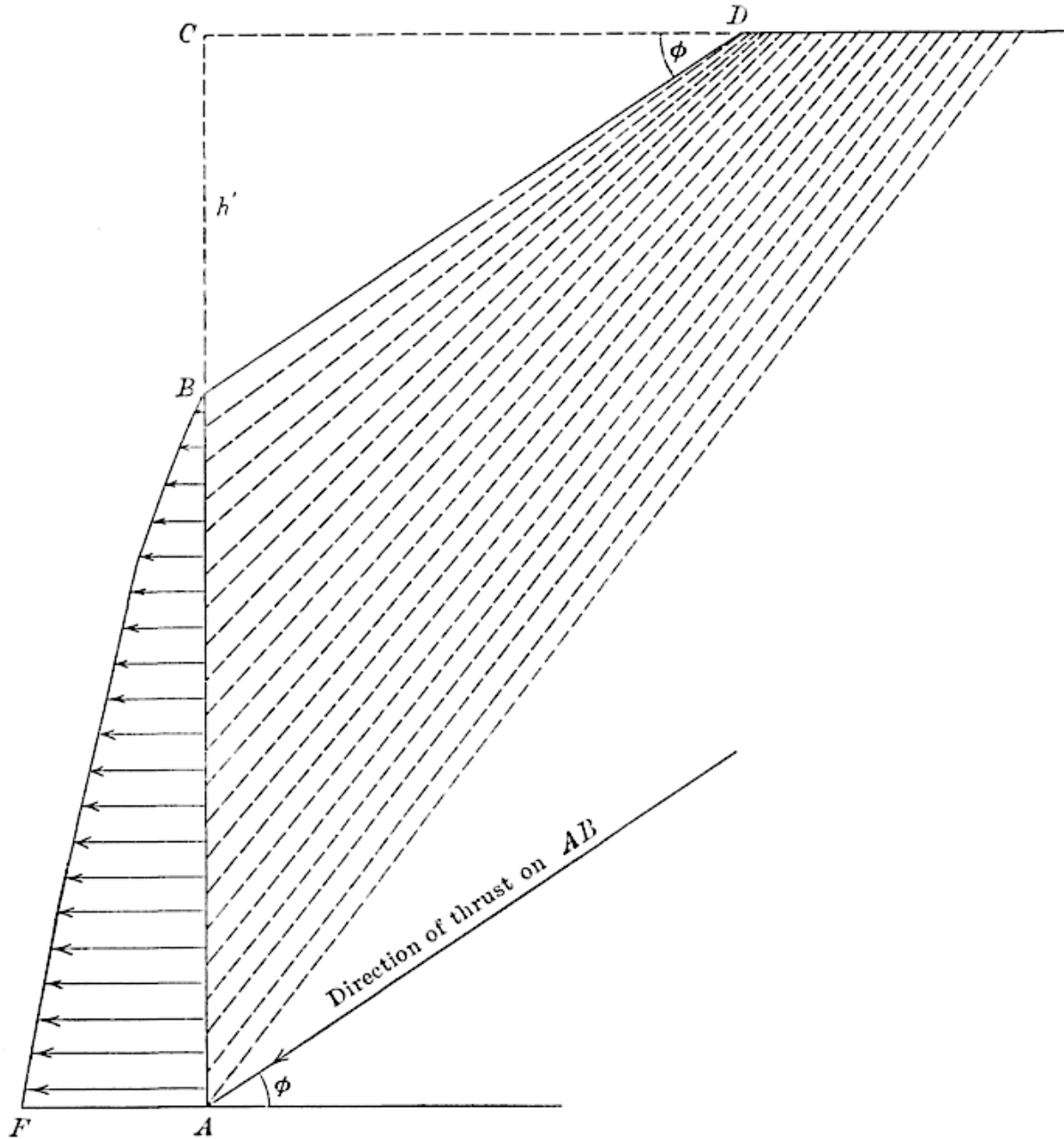


FIG. 10.

As the true position of the center of pressure on a surcharged wall has never been ascertained, as far as the writer knows, he has made a number of constructions, after the method illustrated in [Fig. 1](#), in order to find it.

In place of making the construction for the special case above, it was thought that the results would be more generally useful if the natural slope

was taken with a base of 3 and a rise of 2, and $\phi' = \phi$, therefore $\phi = \phi' = 33^\circ 41'$. The wall, AB , [Fig. 10](#), was taken vertical and 20 ft. high. The surcharge sloped from B at the angle of repose to a point, D , 10 ft. above B , from which point the surface of the earth was horizontal. The face of the wall, AB , was divided into twenty equal parts, 1 ft. each; and, by the construction of [Fig. 1](#), the thrusts (inclined at the angle, ϕ , below the normal to the wall) were found for the successive heights of wall of 1, 2, 3, ... 20, ft., respectively, taking the weight of 1 cu. ft. of earth equal to unity. The successive planes of rupture are shown by the dotted lines in [Fig. 10](#). On the original scale (2 ft. to 1 in.), the upper plane of rupture (for a height of wall = 1 ft.) was found to pass slightly to the right of D .

On subtracting successive thrusts, the thrusts on each foot of wall were obtained. These were plotted as horizontal ordinates at the center of each foot division of the wall, and the “peaks” were slightly rounded off, as shown on the figure. Since, with all care, mistakes amounting to 1% of the total thrusts can easily be made, it was proper to adjust the results in this manner to give the most probable unit pressures on the successive divisions of the wall. The centers of pressure, for heights of the wall varying from 5 to 20 ft., were easily obtained by taking moments about some convenient point; the results are given in [Table 1](#).

Call h the height of wall, measured from B downward, and $h' = BC$, the height of surcharge above the top of the wall; also, let c = the ratio of the distance from the foot of the wall considered to the center of pressure, to the height of the wall. The values of c , for various ratios, $\frac{h'}{h}$, are given in [Table 1](#).

TABLE 1.

$\frac{h'}{h}$	c	$\frac{h'}{h}$	c
∞	0.333	1.00	0.364
...	...	0.75	0.364
2.00	0.353	0.50	0.364
1.50	0.356	...	...
1.25	0.360	0.00	0.333
1.11	0.362		

It is seen, as $\frac{h'}{h}$ diminishes, that c increases, until for $\frac{h'}{h} = 1$, the maximum value for c , 0.364, is attained and remains the same up to $\frac{h'}{h} = 0.50$, after which it probably diminishes, because, for $\frac{h'}{h} = 0$, $c = \frac{1}{3}$.

When some other flatter slope is given to BD , doubtless these values of c will be altered, but, for the case supposed, they should prove serviceable in practice.

Although the earth thrusts on successive portions of AB are really inclined at ϕ° below the normal to AB , they are laid off here at right angles to it, so that the area, ABF , is equal to the total thrust on AB . If the unit pressures varied as the ordinates to the straight line, BF , as for a uniformly sloping earth surface, then, as is well known, $c = \frac{1}{3}$. The area to the left of BF gives the excess thrust which causes c to exceed $\frac{1}{3}$.

Making use of the results of the table as approximately applicable in the foregoing example (Fig. 9), and taking the center of pressure on AF as $0.364AF$ above the base, the resultant there is found to pass 0.02 outside of the base, therefore $q = -0.02$. This experiment on a surcharged wall, of the kind shown, is particularly valuable as being the only one of which any account has been given, as far as the writer knows.

Recurring once more to [Fig. 10](#), it may be recalled that some authors have assumed the unit pressures on AB to vary as the ordinates to a trapezoid, so that the unit pressure at B was not zero (as it should be), but an amount assumed somewhat arbitrarily. In particular, Scheffler derived in this way $c = 0.4$ as an upper practical limit, and used it in making tables for use in practice.

A remark must now be added (relative to all the experimental walls previously mentioned, except Trautwine's), that the friction of the backing on the sides of the box in which the sand was contained has been uniformly neglected. Where the wall is long, this can have little influence, but where the length is not much greater than the height, as in the experiments, this side friction becomes appreciable.

Darwin, as well as Leygue, endeavored to estimate the amount the full thrust (with no side friction) was reduced, by experimenting with sand behind a retaining board, or wall, enclosed in a box as usual, when a partition board was placed perpendicular to the wall and centrally in the mass, and comparing results with those found when the partition board was omitted. Leygue thus found, for walls having a length of twice the height, that the true or full thrust was diminished about 5% from the side friction, for level-topped earth, and as much as 15% for the surface sloping at the angle of repose. If this is true, then the experimental walls just considered would have to be thicker to withstand the actual thrust; or, to put it another way, for the given thickness, the theoretical thrust, including the side friction, would have to be made (as a rough average) about 5% less for the level-topped earth and (roughly) 15% less for the earth sloping at the angle of repose. From the figures it is seen that this will modify the results but slightly, not enough to alter the general conclusion that the theory advocated (including the wall friction) is practically sustained by the experiments, and that the Rankine theory is not thus sustained.

Trautwine's wall consisted of a central portion of uniform height, from which it tapered to the ends, the upper surface being at the angle of repose for the tapered ends. In this case no side friction was developed. The results agree in a general way with the others.

In the many experiments on high grain bins, the enormous influence of the friction of the grain against the vertical walls or sides of the bin has been observed. In fact, the greater part of the weight of grain, even when running out, is sustained by the walls through this side friction. This furnishes another argument for including wall friction in retaining-wall design.

In connection with this subject, it may be observed that many experiments, made to determine the actual lateral pressure of sand or its internal friction angle, are inconclusive, because an unknown part of the vertical pressure applied to the sand in the vertical cylinder or box was sustained by the sides of the cylinder or box. The ratio of lateral to vertical pressures, or the friction angle, cannot be precisely found until the proportion of the load sustained by the sides of the containing vessel has been ascertained experimentally. The writer is of the opinion that the best experiments to aid in the design of retaining walls are those relating to the rotation of retaining walls or boards. The few given herein are the best recorded, though some of them were on models which were too small. In fact, for the small models of Leygue and others, the effect of cohesion is so pronounced that some of the results are very misleading.

As the experiments by Leygue⁴ were very extensive, and evidently made with great care, they will be considered carefully in what follows.

As preliminary to the discussion, however, it is well to give the essentials of Leygue's experimental proof that cohesion and friction exist at the same time. A box without a bottom, about 4 in. square in cross-section and 4 in. high, was made into a little carriage by the addition of four

wheels. The latter ran on the sides of a trough filled with sand which the bottom of the box nearly touched. The box was partly filled with sand, and the trough and box were then inclined at the angle at which motion of the box just began, the sand in the box resting on the sand in the trough, developing friction or cohesion or both, just before motion began. Only friction was exerted after motion began. The solution involves the theory of the inclined plane, but, to explain the principles of the method, it will suffice to suppose the trough and the sand in it to be horizontal, and that the bottomless box filled with sand is just on the point of moving, due to a horizontal force applied to it. The weight of the box and a part of the weight of the sand in it held up by the friction of the sides, is directly supported by the wheels resting on the sides of the trough; so that only a fraction of the weight, W , of the sand in the box is supported directly by the sand in the trough. Call this amount nW . Then, for equilibrium, calling P the horizontal force, less the resistance of the carriage wheels, we have,⁵

$$P = Ak + nWf,$$

where A = the area of the cross-section of the box,

k = the force of cohesion per unit of area

and f = the coefficient of internal friction of the sand.

The value of n was found by weighing: For the dry sand it varied from 0.79 to 0.65, for heights of the sand in the box varying from 1.2 to 3.5 in. For the damp sand and fresh earth (slightly moistened and slightly rammed) which can stand with a vertical face for the height of the box, the filling was loosened by many blows on the box, and n was taken equal to 1.

Three suppositions were made: (1) that both cohesion and friction acted at the same time before motion; (2) that friction alone acted ($k = 0$); (3) that cohesion alone acted ($f = 0$).

The results for various heights of sand in the box are given in [Table 2](#).

TABLE 2.

	(1)		(2)	(3)
	k	f	f	k
Dry sand	7	0.70	0.80 to 0.96	26 to 56
Wet sand	40	0.85	1.20 " 1.90	73 " 133
Very wet sand	31	1.70	2.00 " 2.40	107 " 226
Fresh earth	90	1.63	2.60 " 4.40	150 " 242

The values of k are given in kilogrammes per square meter. It is seen that suppositions (2) and (3) give discordant results, whereas (1), for each kind of filling, gave identical values of k and of f for various heights; hence it may fairly be concluded that, before motion, cohesion and friction both acted at the same time. As to the high values found for k , for the coherent fresh earth, Leygue states that Collin found, by an independent method, for clayey earth and clay of little consistency, $k = 113$ and $k = 193$, respectively. As a further verification of the values of k and f given in (1), it is found that, on using them in the formula for computing the height at which the wet sand or earth will stand vertically, the results agree with experiments.

The values of k , in pounds per square foot, given in Column (1), with the values of ϕ corresponding to the f given, are as follows:

Dry sand, $k = 1.47$, $\phi = 35^\circ$

Wet sand, $k = 8.28$, $\phi = 40^\circ 22'$

Very wet sand, $k = 6.36$, $\phi = 59^\circ 30'$

Damp fresh earth, $k = 18.45$, $\phi = 58^\circ 28'$

It is possible that the method used by Leygue may prove of service to experimenters in obtaining more accurately than hitherto the coefficient of internal friction. Increasing pressures could be obtained by adding weights on top of the sand in the box; but, unless the total weight sustained by friction along the sides of the box is determined carefully for each weight

Suppose Ab_1 to be the plane of rupture, and call the weight of the prism of rupture, ABb_1 , for a thickness of one unit, W . The weight of the prism causes the tendency to slide along the planes, Ab_1 and AB . This tendency is resisted by the reactions of the wall, AB , and the plane, Ab_1 . The reaction of the wall consists of the normal component, E_1 , acting to the right, and the friction resistance, $E_1 \tan \phi'$, acting up. The resultant of these two forces, E , which is equal and opposed to the earth thrust on AB , thus makes the angle, ϕ' , with the normal to the wall. Its direction is given by Af in Fig. 11. The reaction of Ab_1 is made up of the cohesion, C , acting up along Ab_1 , the normal component, N , acting up, and the friction, $N \tan \phi$, acting up along Ab_1 . The two forces, N and $N \tan \phi$ when combined, give a resultant, S , making an angle ϕ with the normal, gs_1 , to Ab_1 . Hence, if the angle, $Igs_1 = \phi$, then gs_1 gives the direction of the resultant, S .

The prism, ABb_1A , is thus in equilibrium under its own weight, W , the cohesive force, C , acting up along Ab_1 , the reaction, E , of AB , acting to the right, and the force, S , acting up. On drawing, to the scale of force, $g1$ vertical and equal to W ; then $1r$ parallel to Ab_1 and equal to C ; then rc_1 parallel to Af to the intersection with gs_1 , the sides of the closed polygon, $g1rc_1g$, in order, will represent the four forces, W , C , E , and S , in equilibrium.

A similar investigation pertains to any other supposed prism of rupture. To find the true one, a number of trial planes of rupture, Ab_1 , Ab_2 , ..., are assumed, and each is treated in turn as a true one (though there can be only one true one). As seen above, the resultant of the normal reaction and friction on any trial plane of rupture must be inclined below the normal to the plane at the angle, ϕ . To lay off the directions of these resultants, from any convenient point, g , say, in the vertical through A , as a center, describe an arc, As_1 , with a convenient radius, gA . With the same radius and A as a center, describe the arc, $Hfd a_1$, cutting the trial planes (produced if necessary) at a_1 , a_2 , ..., a_6 . Let d be the point where the line of natural

slope from A cuts the arc, Ha_1 . On laying off the chords, $As_1, As_2, \dots$, equal to the chords, $da_1, da_2, \dots$, respectively, it will follow that $gs_1, gs_2, \dots$, will make angles, ϕ , below the normals to the planes, $Aa_1, Aa_2, \dots$, respectively. To prove this, take any trial plane, as Ab_1 , which makes the angle, β , with Ad , and drop a perpendicular, gI , from g on Ab_1 (produced if necessary); then, because the sides of the angles are perpendicular, $AgI = Ha_1 = \phi + \beta$, and if $As_1 = \beta$, it follows that $Igs_1 = \phi$, as was to be proved. Hence the chord, $As_1 =$ the chord, $da_1, As_2 = da_2$, etc., as stated.

The weights, in pounds, of the prisms, $ABb_1, ABb_2, \dots$, are, $\frac{1}{2} ep \cdot Bb_1, \frac{1}{2} ep \cdot Bb_2, \dots$, respectively, where p is the length of the perpendicular from A upon Bb_1 (produced if necessary), the foot being the unit of length and e being the weight, in pounds, of 1 cu. ft. of earth. The prisms are supposed to be 1 ft. in length perpendicular to the plane of the paper.

These weights are now laid off to the scale of force, vertically downward from g , to points 1, 2, 3, ..., and from these points, lines are drawn parallel to $Ab_1, Ab_2, Ab_3, \dots$, respectively, of lengths equal to $k \cdot Ab_1, k \cdot Ab_2, \dots$, to represent the forces of cohesion, acting upward along $Ab_1, Ab_2, \dots$, where k = the force of cohesion, in pounds per square foot. From the extremities of these lines, lines are drawn parallel to the direction of the earth thrust on AB (inclined at the angle, ϕ' , below the normal to AB), to the intersections, $c_1, c_2, c_3, \dots$, with $gs_1, gs_2, gs_3, \dots$, respectively. With dividers, it is found, for this figure, that nc_4 is the longest of these lines; whence nc_4 , to the scale of force, measures the earth thrust against AB , in pounds. This follows, because, for any less thrust, since n is fixed, when nc_4 becomes less, c_4 falls to the left of the first position, and the new gc_4 , representing the thrust on Ab_4 , due to the normal reaction on it and friction only, will make a greater angle than ϕ to the normal to Ab_4 , which is inconsistent with the laws of stability of a

granular mass. In fact, if N is the normal component of the thrust on the plane, Ab_4 , $N \tan \phi$ is all the friction that can be exerted on it. The resultant of N and $N \tan \phi$ thus makes an angle, ϕ , with N , and this angle cannot be exceeded. The true thrust on the wall, AB , is thus the greatest of the trial thrusts, nc_4 . The prism of rupture, Ab_4B , is in equilibrium under the four forces represented by the sides of the closed polygon, $g4nc_4g$; $g4$, representing its weight; $4n$, the cohesion acting along Ab_4 ; nc_4 , the reaction of AB (opposed and equal to the earth thrust); and c_4g the reaction of the plane, Ab_4 , due to the normal component and friction on it only. The full reaction of the plane can be found by combining the forces, given in magnitude and direction by $4n$ and c_4g , but it is not needed.

It is to be noted that nc_4 is the least thrust for which equilibrium is possible. The other trial thrusts should now be lengthened to equal nc_4 , since this is the true thrust or reaction of the wall, AB . All the new points, c_1, c_2, c_3, c_5, c_6 , will now lie to the right of the old points; hence the new $gc_1, gc_2, gc_3, gc_5, gc_6$, will all make less angles than ϕ with the normals to the planes, Ab_4 , etc.; hence stability everywhere in the earth mass is assured.

The solution represented by Fig. 11 is general, and applies whether AB is inclined to the right or left of the vertical through A or coincides with it, and whether the earth surface Bb_6 is horizontal or inclined above or below the horizontal. It can likewise be easily adapted to the case shown in Fig. 1.7 The construction of Fig. 11 has been used in evaluating the thrust and determining (approximately) the plane of rupture in the experiments (recorded below) of Leygue on retaining boards, AB , that could be rotated about A , and thus

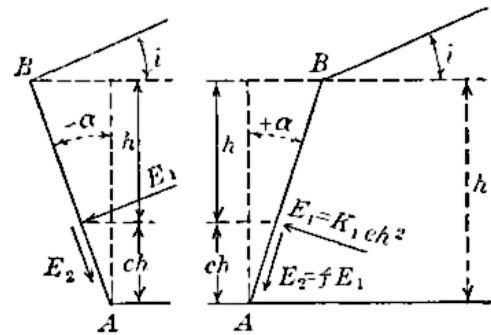


FIG. 12.

placed at any inclination to the vertical. In all the experiments, the vertical height of AB was 0.656 ft.; the length of the board was 1.3 ft. The value of the moment of the earth thrust about A was found by use of suitable apparatus, corresponding to dry sand with a natural slope of 3 base to 2 rise, or $\phi = 33^\circ 41'$, $\phi' = \phi$, and $c = 89$ lb. per cu. ft. By use of the partition board mentioned previously, the side friction of the sand on the glass sides of the box containing it was estimated, and the moments corrected, so as to give the true moment when there is no side friction. The notation used to express results is partly given in Fig. 12, for the general case where,

- h = vertical height of the board, AB ;
- ch = vertical height to center of pressure on AB ;
- α = angle AB makes with the vertical, counted as positive, or negative according as AB lies to the right or left of the vertical through A ;
- i = inclination of surface of earth to the horizontal;
- $E_1 = K_1 e h^2$ = normal component of thrust on AB acting ch feet vertically above A ;
- $E_2 = f E_1 = \tan \phi' E_1$ = component of thrust acting in the direction from B to A , $f = \tan \phi'$;
- γ = angle plane of rupture makes with horizontal.

The resultant, E , of E_1 and E_2 , evidently makes the angle, ϕ' , with the normal to AB . The moment of this resultant about $A = M = E_1 h \sec \alpha = c K_1 \sec \alpha \cdot e h^3 = m e h^3$, if we put $m = c K_1 \sec \alpha$. From the last formula, it is seen that m is the moment of the thrust about A , for $e = h = 1$; also from $E_1 = K_1 e h^2$, it follows that K_1 is the normal component of the thrust for $e = h = 1$.

When cohesion is included, c is not exactly $\frac{1}{3}$, but it is very slightly less for $k = 1$ or 2. It will be assumed at $\frac{1}{3}$, and, from the values of m given by Leygue, K_1 will be derived from the formula above,

$$K_1 = 3 m \cos \alpha.$$

Thus, for the case represented by Fig. 11, $\tan \alpha = \frac{1}{3}$, therefore $\alpha = 18^\circ 26'$, $\cos \alpha = 0.949$; $\tan i = \frac{2}{3}$. Experiment gave $m = 0.032$, therefore $K_1 = 3 \times 0.032 \times 0.949 = 0.091$.

Neglecting cohesion, theory gives $K_1 = 0.182$, or twice the amount given by experiment. If, however, the construction of Fig. 11 is made for the actual height of the retaining board, $h = 0.656$ ft., $e = 80$, $k = 1$ (cohesion, in pounds per square foot), we find $E_1 = 2.75$. On substituting this in the formula, $E_1 = K_1 e h^2$, we have,

$$2.75 = 80 (0.656)^2 K_1, \text{ therefore } K_1 = 0.080.$$

By a comparison of the values, it is evident that, if the cohesion was assumed at a little less than 1 lb. per sq. ft., the theoretical and experimental values could be made to agree exactly. The case just examined exhibits the most pronounced difference between the ordinary theory (corresponding to $k = 0$) and experiment, of any shown in Table 3. Further, it will be observed, that, for an assumed cohesion of about 1 lb. per sq. ft., the theoretical and experimental values for all the cases given by Leygue very nearly agree.

The value, $e = 80$, in place of Leygue's, $e = 89$, was used, which would alter the results somewhat, but not the general conclusions. The construction of Fig. 11 will give E and its normal component, E_1 , with practical accuracy, but it is not readily adaptable in finding the plane of rupture. In most of the drawings a small scale was used, in order to limit the drawing to a sheet of writing paper, hence, on both accounts, γ cannot be counted on to nearer than 1° or 2° , except for $k = 0$, when γ was found by computation, or by the construction of Fig. 1.

TABLE 3.

Tan α .	Tan i .	Cohesion, k , in pounds per square foot.	ANGLE OF RUPTURE WITH THE HORIZONTAL, γ .		COEFFICIENT K_1 OF THE NORMAL COMPONENT OF THE THRUST $E_1 = K_1 e h^2$.	
			Theory.	Experiment.	Theory.	Experiment.
$+\frac{1}{3}$	0	0	50° 12'	51° 30'	0.060	0.043
		1	51°		0.042	
		2	52° 30'		0.026	
$+\frac{1}{3}$	$\frac{2}{3}$	0	33° 41'	47°	0.182	0.091
		1	44°		0.084	
		2	49°		0.043	
0	0	0	56° 36'	56° 30'	0.111	0.090
		1	57°		0.093	
		2	58°		0.077	
		3	58° 30'		0.062	
0	$\frac{1}{2}$	0	47° 30'	51°	0.178	0.141
		1	50°		0.148	
		2	53°		0.121	
		3	55°		0.098	
0	$\frac{2}{3}$	0	33° 41'	49°	0.345	0.195
		1	44°		0.205	
		2	46° 30'		0.150	
		3	50°		0.111	
$-\frac{1}{3}$	0	0	60° 21'	61°	0.185	0.179
		1	63°		0.171	
		2	63°		0.155	
$-\frac{1}{3}$	$\frac{2}{3}$	0	33° 41'	57°	0.660	0.387
		1	57°		0.267	
		2	50°		0.236	

The results in [Table 3](#) are remarkable, and explain quite satisfactorily how Leygue, Darwin, and others found, by experiments on small models, results differing so much from the ordinary theory, where cohesion is neglected.

It should be remarked that the values of γ given in [Table 3](#) under “Experiment,” are not exactly those given by Leygue in his tables, but are the averages obtained from the two sets of drawings given by him in the plates, and represent the inclinations of the chords of the really curved surfaces of rupture. His experiments with the spring apparatus are not considered, as the results are open to doubt, because the prism of rupture, in descending, could not slide down freely, but as it advanced would rub over the floor, thus lessening the thrust there considerably.

From [Table 3](#), the results given by experiment are seen to differ widely from the ordinary theory in which $k = 0$.

The discrepancies are largely, or almost entirely, due to the very small models used, as will be evident from the following considerations: Suppose the height, h , of the wall, AB , to be 10 times the height given in [Fig. 11](#), or 6.56 ft.; then, as the areas of triangles such as $AB b_1$, etc., vary as the squares of the heights, but the lengths of sides, as $A b_1$ etc., vary only as the first power of the heights, the weights of the successive trial prisms of rupture will be 10^2 or 100 times as great as before, whereas the corresponding cohesive forces, acting along the planes, $A b_1$, etc., will be only 10 times the first values. Hence, if we use a scale of force $\frac{1}{100}$ of the former scale, the weights of the prisms, $AB b_1$, $AB b_2$, etc., will be represented, as before, by $g\ 1$, $g\ 2$, etc., but the lines representing the cohesive forces will be only $\frac{1}{10}$ of the former lengths. Thus the new $4\ n$, [Fig. 11](#), will be laid off from 4 only $\frac{1}{10}$ of the length shown in the figure.

The relative decrease in the lines representing cohesive forces will be still more marked for a wall $20 \times 0.656 = 13.12$ ft. high, the weights of

prisms being 400 times as great, but the cohesive forces only 20 times as great as before. It is evident from this reasoning that, for $k = 1$, the cohesive forces are practically negligible for walls, say, 10 ft. high, especially if the earth surface is level. In fact, a little examination of the original drawings showed, for walls about 6 ft. high, that the earth thrust, neglecting cohesion, was only from 1 to 5% in excess over that for $k = 1$. The smaller percentages referring to $\tan \alpha = -\frac{1}{3}$, or $\tan \alpha = 0$, while the larger percentages referred to $\tan \alpha = +\frac{1}{3}$, for earth surface horizontal or sloping at the angle of repose.

Such results should be of great service to future experimenters as proving two things: (1) that dry sand, with as small a coefficient of cohesion as possible, should be used (perhaps grain would be a more suitable material), and (2) that no experimental wall should be less than from 6 to 10 ft. high.

Even if the wall is, say, 6 ft. high, if damp clayey earth is used as the filling, with a coefficient of adhesion, $k = 10$, then all the diagrams of forces, as in [Fig. 11](#), will be the same as before, or similar figures, and the discrepancies noted in [Table 3](#), will be as pronounced as ever. All the experiments on retaining boards, except some of Curie's, have been with very small models, and the results have brought the common theory under suspicion, if not into disrepute.

The writer hopes that the foregoing investigation and results may be instrumental in establishing more confidence in the theory, and in showing when cohesive forces may be practically neglected and when they must be included.

As an illustration, the results for a vertical wall 10 ft. high are presented in [Table 4](#), taking $\phi = 33^\circ 41'$ and $e = 80$. In the first wall, the surface of the earth was horizontal; in the second wall its slope was 1 rise to 2 base.

TABLE 4.

Tan α .	Tan i .	k .	γ .	K_1 .
0	0	0	56°36'	0.111
		1	56°	0.110
		5	57°	0.101
		10	57°	0.096
1	$\frac{1}{2}$	0	47°30'	0.178
		1	49°	0.176
		5	49°	0.165
		10	49°	0.155

In [Table 4](#) the results for $k = 0$ and $k = 1$ are practically the same, but K_1 for $k = 10$ is 13% less than for $k = 0$. If the value, $k = 18.5$, for fresh earth slightly damp and lightly rammed, given by Leygue above, is even approximately correct, it is seen that, for such a filling, the effect of cohesion must be included to get results at all agreeable with experience or experiment.

Recurring to the experimental retaining walls proper, [Figs. 4 to 9](#), it is evident from the foregoing, that cohesion will affect the results inappreciably, except perhaps in the case of [Figs. 6 and 7](#), where the height was about 0.6 ft. Assuming $k = 1$, it seems to be probable, from the results of [Table 4](#), that the thrust should be decreased in the ratio of 93:111. Effecting the construction for the new thrust, it is found that the point, I, falls within the base, 0.03 of its width for [Fig. 6](#) (Trautwine's wall), and 0.02 of its width for [Fig. 7](#) (Curie's wall).

The theory advocated is thus practically sustained by all the experiments given above, either on retaining boards or retaining walls proper, when a coefficient of cohesion of about $k = 1$ for dry sand is used.

The method of evaluating the thrust, given in [Fig. 11](#), is as valid when $k = 0$, or cohesion is neglected, as in the ordinary theory. The lines parallel to the thrust are now drawn directly from 1, 2, ..., to the intersection with

$g c_1, g c_2, \dots$, and the greatest one is taken for the true thrust. Although the writer expressly disclaims any great accuracy in the values of K_1 in [Table 4](#), on account of the small scale of the drawings, nevertheless, the results by the construction for $k = 0$ and $\tan i = 0$ or $\frac{1}{2}$, were found to differ from computed values only 2, 3, 0, and 1% for the different cases, which should give confidence in the general conclusions, at least.

The diagram, [Fig. 11](#), with a slight modification, can be utilized to find the coefficient of cohesion, k , at which the bank of earth will stand without a retaining board. Thus, let each line, as $4 n$, representing the cohesive force acting along its proper plane, be extended to meet the corresponding $g c$; any such line measured to the scale of force and then divided by the length of the plane along which it acts, will give the cohesive force, in pounds per square foot, corresponding to no thrust on AB , for the particular plane considered. The greatest of these values is evidently the value of k for which the filling will stand without a retaining board. The work can be much abbreviated by using a well-known principle, that the plane along which the unit cohesion is greatest (the plane of rupture) bisects the angle, $BA d$, between the surface, AB , and the line of natural slope. Suppose $A b_1$ to be this plane, then we have only to extend $1 m$ to meet $g c_1$, at 0, measure 10 to the scale of force, and divide by the length of $A b_1$, to the scale of distance, to find the coefficient desired. By either method it was found that a cohesive force of 7 lb. per sq. ft. was required to sustain a mass of earth with a vertical face, $AB = 0.656$ ft. high, when $B b_6$ was horizontal.

It was stated, in connection with [Equations \(1\) and \(2\)](#), referring to the thrust on a vertical wall of height, h , with the earth surface sloping at the angle of repose, that this particular case would be discussed later. To show the influence of cohesion, the planes of rupture for such a wall, 2.4 ft. high, for various values of k (in pounds per square foot), are given in [Fig. 13](#). The values of K_1 (for $e = 80$ and $\phi = 33^\circ 41'$) are as follows:

$k = 0 \dots K_1 = 0.345$	$k = 5 \dots K_1 = 0.183$
$k = 2 \dots K_1 = 0.237$	$k = 10 \dots K_1 = 0.120$
$k = 3 \dots K_1 = 0.215$	$k = 15 \dots K_1 = 0.075$

The first value was found by computation, the others by construction. As is well known, the theoretical plane of rupture approaches indefinitely the natural slope as k approaches zero. For appreciable cohesion (and there is always some cohesion) the plane of rupture lies above the natural slope, with very materially decreasing normal components to the thrust as k increases. As the height of wall increases, the influence of cohesion

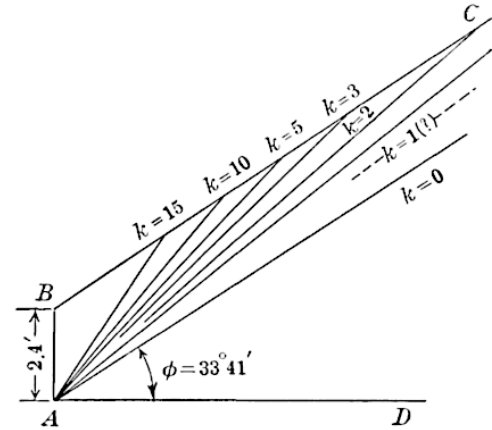


FIG. 13.

diminishes. Thus, as shown above, for a wall 5 times 2.4 ft., or 12 ft. high, the weights of the prisms, ABC , etc., are 25 times the former values, but the cohesive forces, which vary directly as AC , etc., are only 5 times the first values. Hence, if the former values of k are multiplied by 5, the new diagram of forces, Fig. 11, will be similar to the old one. Thus, for the wall 12 ft. high, the plane of rupture and the value of K_1 , for $k = 10$, correspond to the old values for $k = 2$, for $k = 15$, to the old values for $k = 3$. For fresh earth filling, slightly packed, it is possible that the values, $k = 10$, $k = 15$, may be reached, with a material reduction in K_1 from the values given by Equations (1) and (2). As the height of the wall increases, say to 25 or 50 ft., the influence of cohesion, in diminishing the thrust, becomes very small, and it is better to ignore it altogether. In fact, as we know very little, and that imperfectly, of the coefficients of cohesion, it is perhaps safer, at present, to use Equations (1) and (2) in all cases. It is very evident, though, that for most cases in practice, the formulas give a very appreciable excess over the true thrust, and that the true plane of rupture never coincides with the natural slope.

From all that precedes, it is seen that the results of experiments on small models in the past have proved to be very misleading, and that experiments on large models are desirable, and can alone give confidence. Leygue has made such experiments on retaining boards, from 1 to 2 m. (3.28 to 6.56 ft.) in height, simply to determine the surface of rupture. This is really the essential thing, for, as soon as the prism of rupture is known, the thrust is easily found. In a general way, the results agree with theory when the cohesion is neglected, though the curved surfaces of rupture were very irregular, particularly for the stone filling. The first two experiments were made with both dry and damp sand as a filling; the next six, with stones varying from 1.5 to 20 in. in diameter. In another series of five experiments, sand was used. In all the foregoing experiments, the surface of the material was horizontal. In three additional experiments, the walls were surcharged with sand as a filling. In one experiment, the wall was 6.56 ft. high and the surcharge was 3.28 ft.; in another experiment, the wall was 3.28 ft. high, and the sand, sloping from its top at the angle of repose, as in the former case, extended to 3.28 ft. above the wall, where the surface was horizontal.

Applying the construction of [Fig. 1](#), it was found that the plane of rupture passed, say, 2° above that given by experiment in the first case and about 3° below in the second. It will be evident from the construction of [Fig. 11](#), omitting cohesion, that trial planes of rupture differing by 2 or 3° from the true one, give nearly the same thrust. Taking the average, these experiments on large models, tend, in a general way, to sustain the theory.

In a paper by the late Sir Benjamin Baker, Hon. M. Am. Soc. C. E., "The Actual Lateral Pressure of Earthwork,"⁸ two experiments by Lieut. Hope and one by Col. Michon, on counterforted walls, are given. Although such walls do not admit of precise computation, on account of the unknown weight of earth carried by the counterforts, through friction caused by the thrust of the earth in a direction perpendicular to the counterforts, still the computation was made, as the conclusions are interesting. Therefore, the

first vertical wall of Lieut. Hope was examined, especially as Mr. Baker, using the Rankine theory, found, for this wall, the greatest divergence between the actual and the Rankine thrust, of any retaining wall examined.

At the moment of failure, the wall was 12 ft. 10 in. high, the thickness of the panel was 18 in., and the counterforts were 10 ft. from center to center, projecting 27 in. from the wall, or 3 ft. 9 in. from the face, as inferred from the next example. As it is stated that the wall had the same volume as the 10-ft. wall previously examined in this paper (Fig. 4), the counterforts must have been 2 ft. thick. Assuming these dimensions, and using the values given; $\tan \theta = \frac{3}{4}$, or $\theta = 36^{\circ} 52'$ (say $= \phi'$), $e = 95.5$ lb. = weight of a cubic foot of earth, and $w = 100$ lb. = weight of a cubic foot of masonry, we first compute $E_1 = 1\,496$ lb., the normal component of the earth thrust on a length of 1 ft. of wall. The normal thrust on the panel is thus $8 E_1$ and on the counterfort $2 E_1$. The friction (acting vertically downward) caused by this thrust is $8 E_1 \tan \theta$ on the panel and $2 E_1 \tan \theta$ on the counterfort. The moment of these forces about the outer toe of the wall, totals 39 800 ft-lb. The resisting moment of 10 ft. in length of combined panel and counterfort, about the outer toe, assuming the wall to be vertical, is 29 800 ft-lb. If, to the latter, we add the moment of 17% of the weight of earth between the counterforts, supposed to be held up by the sides of the latter, the total moment exactly equals the first. However, at the moment of failure by overturning, the panels had bulged $4\frac{1}{2}$ in. and the overhang at the top was $7\frac{1}{2}$ in. Taking the moment of stability of the wall at 26 000 ft-lb. (Mr. Baker's figure), it is found that, for equilibrium, 24% of the weight of earth between the counterforts must be carried by them, When the earth was 8 ft. high, a heavy rain was recorded, so that, doubtless, some appreciable cohesion was exerted, though necessarily omitted in the computation.

The experimental wall of Col. Michon was 40 ft. high, with very deep counterforts, only 5 ft. from center to center. The very heavy and wet filling

between the counterforts, being treated as a part of the wall, a construction (made on the printed drawing) shows that the resultant of earth thrust and weight of wall passes through the outer toe. Doubtless the cohesion factor in this wall was large. In the paper mentioned, the details as to Gen. Burgoyne's experimental walls are given. There were four of these walls, each 20 ft. long, 20 ft. high, and with a mean thickness of 3 ft. 4 in. Two of the walls were perfectly stable, as in fact theory indicates for all four walls if they were monolithic. The other two walls fell, one bursting out at 5 ft. 6 in. from the base, and the other (a vertical wall), breaking across, as it were, at about one-fourth of its height. As these walls consisted of rough granite blocks laid dry, it is highly probable that the breaks were due to sliding, owing to the imperfect construction; besides, "the filling was of loose earth filled in at random without ramming or other precautions during a very wet winter."

From a consideration of all the observations and experiments (some of them unintentional), Mr. Baker concludes that the theoretical thrust is often double the actual lateral pressure. He used the old theory, which neglects both cohesion and wall friction. If he had included them, the resulting theory would not have been so deficient "in the most vital elements existent in fact" as he charges against the "textbook" theory.

However, the writer must be clearly understood as not recommending that cohesive forces be considered in designing a retaining wall backed by a granular material, such as fresh earth, sand, gravel, or ballast. It has been the main object of this paper to show that, although cohesive forces must be included in interpreting properly the results on small models and many retaining walls, yet, for walls more than 6 or 10 ft. in height, backed with dry fresh material, not consolidated, the cohesive forces can be practically neglected in design. Hence, experimenters are strongly advised to leave small models severely alone and confine their experiments to walls from 6 to 10 ft. high, backed by a truly granular material, such as dry sand, coal,

grain, gravel, or ballast, where the cohesive forces will not affect the results materially. Further, it is evident that walls of brick in wet sand, or walls of granite blocks, etc., laid dry, are very imperfect walls. The overhang, just before falling, is large, and the base is often imperfect. For precise measurements, a light but strong timber wall on a firm foundation, seems to be best; and the triangular frame of [Fig. 8](#) seems to meet the required conditions very well, especially if the framing is an open one, with a retaining board only on one leg. The base thus becomes wider, and the overhang less, than with any rectangular wall.

When the design of a wall to sustain the pressure of consolidated earth is in question, even if a perfect mathematical theory existed, it would still prove of little or no practical value, because the coefficients of friction and cohesion are unknown. The coefficient of friction at the surface can be easily found, but it is a difficult matter to find the coefficient of cohesion, which doubtless varies greatly throughout the mass.

Mr. W. Airy, in his discussion of Mr. Baker's paper, states that he found the tensile strength of a block of ordinary brick clay to be 168 and of a certain shaley clay 800 lb. per sq. ft., the coefficients of friction for the two materials being 1.15 and 0.36, respectively. Cohesive resistance is more analogous to shear, but such figures indicate the wide variations to be expected, particularly in k , the coefficient of cohesion. If this coefficient is to be guessed at, in order to substitute it in the supposed perfect formula, then it is plainly better to guess at the thickness of the wall in the first instance.

As an illustration, consider the well-known equation:⁹

$$h = \frac{4k}{e} \frac{1 + \sin \phi}{\cos \phi} = \frac{4k}{e} \left(f + \sqrt{1 + f^2} \right)$$

which gives the height, h , of vertical trench that will stand without any sheeting.

In this equation,

k = the cohesive or shearing resistance, in pounds
per square foot;

e = the weight of a cubic foot of earth, in pounds

$f = \tan \phi$;

ϕ = the inclination of a natural slope; and

h = the height of the unsupported bank, in feet

Thus, if $f = 0.7$, whence $\phi = 35^\circ$, the equation reduces to

$$h = 7.68 \frac{k}{e}.$$

For, $e = 100, k = 100, h = 7.68$ ft.,

$e = 100, k = 200, h = 15.36$ ft.,

$e = 100, k = 300, h = 23.04$ ft.

As certain trenches with vertical sides have been observed to stand unsupported for heights of 15 or even 25 ft., the equation would seem to indicate that cohesive or shearing resistances of about 200 to 300 lb. per sq. ft. were required to cause equilibrium. If friction is not supposed to be exerted, then $f = 0$ and $h = 4 \frac{k}{e}$; and, for the same unsupported heights, the cohesion would be about doubled. Evidently, if cohesion, which (to judge from Mr. Airy's experiments) may vary from one to several hundred pounds per square foot, has to be guessed at in order to determine h , it is plainly better to guess at h at once.

The foregoing equation cannot be regarded as giving very accurate results, mainly because a plane surface of rupture is assumed, whereas, from both theory and observation, this surface is known to be very much curved; besides, the cohesion and friction along the ends of the break have been neglected. However, the hypothesis of a plane surface of rupture, the ends being supposed to be included, gives a greater value to h than the true one, whereas, neglecting the influence of the ends, it tends in the other direction; so that the equation may not err so greatly.

In the discussion of the paper¹⁰ by J. C. Meem, M. Am. Soc. C. E., E. G. Haines, M. Am. Soc. C. E., states that where breaks occur in the sides of an unsupported trench, the solid of rupture often approximates to a quarter sphere, surmounted by a half-cylinder of the same height, the radii of the sphere and cylinder being equal. In Fig. 14, let $AIBE$ represent the quarter-sphere, ABG the half-cylinder, and $EACDB$ the face of the trench. According to the observations of Mr. Haines, when the part, $ACDB$, of the side of the trench is supported by sheeting and bracing, it sometimes happens that a part of the quarter-sphere, $AIBE$, breaks out, so that the semi-cylinder above would descend but for the bracing, the thrust of which, it is supposed, induces arch action in the earth.

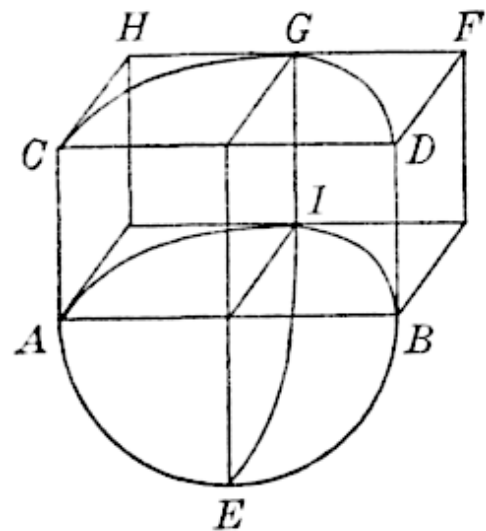


FIG. 14.

This is possible; but, if so, as the sheeting is not supposed to be carried to the bottom of the trench, there can be no vertical component in its reaction, and the thrust, B , of the braces and sheeting, acting on $ACDB$, must be horizontal; further, the earth cannot act as a series of independent

arches devoid of frictional resistance between them, but must act as a whole.

Another way of explaining the phenomena is to suppose the horizontal thrust of the braces, B , on the exposed face, $ABCD$, to cause friction at the back of the break of sufficient intensity to prevent the semi-cylinder from descending, just as a book can be held against a vertical wall by a horizontal push.

To illustrate the principle, it will suffice to replace the semi-cylinder by the circumscribing parallelopiped, AF , and suppose it to be held up by the friction on the back face, with possibly cohesion acting on the three interior vertical faces. Thus, let $AC = CH = a$, and $AB = 2a$; then the friction on the back face is Bf , the cohesion on the three faces is $(2a^2 + 2a^2)k$, and the weight of earth, AF , equals $2a^3e$. Hence, as friction and cohesion always act opposite to the incipient motion, or vertically upward in this case,

$$Bf + 4a^2k = 2a^3e.$$

Evidently, the value of B , derived from this equation, gives an extreme upper limit, which is doubtless never attained, as there is nearly always some support from the earth which has not broken out below the level of ABI .

Where the sheeting and bracing are of sufficient size, are tightly keyed up, and extend to the bottom of the trench, or where the bank is supported by a retaining wall, the earth near the bottom cannot break out, and the equation is not valid.

However, if, from any cause, such as insufficient sheeting, the break has taken place over even a part of $AIBE$, the mass, AF , above will tend to tip over at the top, giving the greatest pressure on the top braces. This appears to explain the phenomena observed by Mr. Meem and others in connection with some trenches.

With regard to tunnel linings, as is well known, the vertical pressure on the top is generally small, the great mass of earth vertically over the tunnel being largely held up by the friction of the earth (caused by the earth thrust) on its vertical sides, exactly as in the case of tall bins, where most of the weight of the grain is held up by the sides of the bin, the theory being very similar in the two cases. In consolidated earth, cohesion assists very materially in this action.

It might be inferred, from the facts of observation, that consolidated earth acts as a solid, though, of course, it differs from a solid in this: that its physical constants (cohesion, friction, etc.) vary enormously with the degree of moisture. It is likely that these constants alter with the depth, and likewise are subject to changes from shocks.

It is a question too, whether, as is the case with loosely granular materials, friction acts (before rupture) at the same time with shear or cohesion in consolidated earth. From the interesting remarks¹¹ of Mansfield Merriman, M. Am. Soc. C. E., on internal friction, it seems probable that friction and shear exist at the same time in a solid; but, to reach sound conclusions, as he states, “further studies on internal friction and on internal molecular forces are absolutely necessary.”

From the present state of our knowledge with respect to the theory and physical constants pertaining to consolidated earth, it would seem that experience must largely be the guide in dealing with it. The facts are supreme—the rational theory may come later.

Similarly, for retaining walls backed by loosely aggregated, granular materials, the facts are supreme, and, on that account, they have been presented very fully in this paper; further, a theory has been found to interpret them properly. It is true that the fresh earth, from the time that it is deposited behind a retaining wall, begins to change to a consolidated earth, from the action of rains, the compression due to gravity, and the influence

of those cohesive and chemical affinities which manufacture solid earths and clays out of loosely aggregated materials, and even cause the backing sometimes to shrink away from the wall intended to support it; but it is plain that the wall should be designed for the greatest thrust that can come on it at any time, and this, in the great majority of cases, will occur when the earth has been recently deposited.

The cases which have been observed where the bank has shrunk away from the wall and afterward ruptured (after saturation, perhaps) are too few in number to warrant including in a general scheme of design, even supposing that a rational theory existed for such cases. A few remarks on the theory pertaining to the design of retaining walls may not be inappropriate. From the discussion of all the experiments referred to in this paper, the conclusion may be fairly drawn that the sliding wedge theory, involving wall friction, is a practical one for granular materials of any kind subjected to a static load. In practical design, however, vibration due to a moving load has to be allowed for; also the effect of heavy rains. Both these influences tend generally to lower the coefficients of friction and add to the weight of the filling. Mr. Baker says:

“Granite blocks, which will start on nothing flatter than 1.4 to 1, will continue in motion on an incline of 2.2 to 1,¹² and, for similar reasons, earthwork will assume a flatter slope and exert a greater lateral pressure under vibration than when at rest.”

Instances of slips in railway cuttings, caused by the vibration set up by passing trains, have been given by many engineers. The effect of vibration is most pronounced near the top of a retaining wall, and is evidently greater for a low wall than for a high one. All the influences cited can only be included under the factor of safety, and the writer recommends for walls from 10 to 20 ft. in height a factor of 3. This may be increased to 3.5 for walls 6 ft. high and decreased to 2.5 for walls 50 ft. high, or those with very high surcharges. In the application, the normal component of the earth

thrust on the wall, E_1 , will alone be multiplied by the factor, the friction, $E_1 \tan \phi'$, exerted downward along the back of the wall, being unchanged. This allows very materially for a decrease in ϕ' due to rains and vibration, as well as for an increase in the thrust, due to ϕ becoming less.

The effect is illustrated in Fig. 15, where a retaining wall is supposed to be subjected to the earth thrust, CF , making an angle ϕ' with the normal to the face, AC , of the wall. The component of CF normal to AC is $E_1 = CD$, the component acting downward along CA is represented in magnitude and direction by DF , which equals $E_1 \tan \phi'$. Suppose the factor of safety to be 3, then CD is extended to H , making $CH = 3CD$; HG

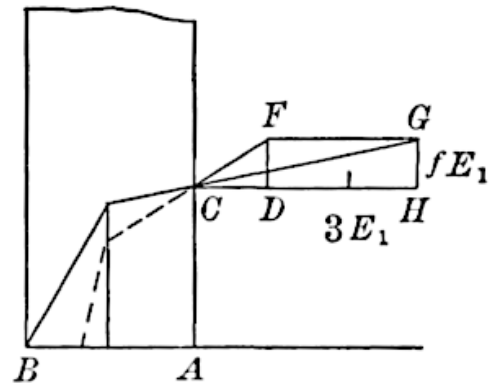


FIG. 15.

is drawn equal and parallel to DF ; whence GC will represent the thrust, which, combined with the weight of the wall, acting through its center of gravity, must pass through the outer toe of the wall.

To see what thickness of a vertical rectangular wall corresponds to this factor of safety, 3, for $\phi = \phi' = 33^\circ 41'$, or a natural slope of 3 base to 2 rise, let it be assumed that the weights per cubic foot of earth and cut-stone masonry in mortar are in the ratio of 2:3; then, for level-topped earth, a computation shows that, for the factor, 3, the base of the wall must be $0.34 h$. If the earth slopes indefinitely at the angle of repose from the top of the back of the wall, and a factor 2.5 is used, then the thickness will be $0.49 h$.

For brick masonry in mortar, the specific weight of which is $\frac{5}{4}$ of that of the filling, the foregoing thickness would be changed to $0.37 h$ and $0.52 h$, respectively, h being equal to the height of the wall.

It must be noted especially, however, that if the original earth thrust, when combined as usual with the weight of wall, gives a resultant which passes outside of the middle third of the base of the wall as computed above, then the thickness must be increased, so that the resultant will at least pass through the outer middle-third limit. This ensures compression over the whole base and no opening of part of the joint under normal conditions. With regard to the thickness above of about one-third of the height, Mr. Baker states that hundreds of brick revetments have been built by the Royal Engineer officers, with a thickness of only $0.32 h$ for a vertical wall. He advises, as the result of his own extensive experience, that the thickness be made one-third of the height for level-topped earth of average character, and that the wall be battered $1\frac{1}{2}$ in. to the foot. He states, further, that, under no ordinary conditions of surcharge on heavy backing is it necessary to make the thickness of a retaining wall on a solid foundation more than one-half the height. The thicknesses computed above agree fairly well with those recommended by Mr. Baker, and it would seem that a table of thicknesses computed on the above basis should correspond to safe walls under ordinary conditions.

It has been noted above that [Equation \(1\)](#), corresponding to a slope of indefinite extent, probably gives too great a thrust; besides, there are no embankments with such a slope. An embankment from 100 to 150 ft. high, supported by a low wall, may approximate the conditions assumed, but, before it is finished, the earth has consolidated to such an extent that the actual thrust is doubtless much less than the computed one. The truth is that, in nearly all back-filling of ordinary earth, the cohesive and chemical affinities commence their work very soon after the filling is deposited, and consolidation is gradually effected; so that, as has been stated, the actual thrust is often much less than is estimated in the design of the wall, where cohesive forces are neglected. In many old walls, as has been observed, the consolidation has gone so far that the backing has shrunk away from the wall altogether. It would be hazardous, though, to allow for cohesion, in a

wall backed by fresh earth, unless the surcharge was high and was a long time in building. Finally, it should be observed that the footing of a retaining wall should be wide, and should always be tilted at such an angle that sliding is impossible.

A glance at [Figs. 4, 5, and 6](#), will make it apparent that the Rankine and other theories differ in their results mainly because of the assumed difference of inclination of the earth thrust. In the design of walls, however, the method proposed ([Fig. 15](#)) will approximate in results those given by the Rankine theory, where, say, the earth thrust, whether inclined or not, is multiplied by the factor of safety. The writer does not advocate the middle-third limit method in design, as it gives variable factors of safety for different types of walls. Besides, if the actual resultant on the base passes one-third of its width from the outer toe, there is no pressure at the inner toe, and the unit pressure at the outer toe is double the average. If vibration or other cause increases the thrust, the joint at the inner toe opens, and the pressure is concentrated too much near the outer toe. In the reinforced concrete wall, the earth thrust on a vertical plane through the inner toe is required. As this plane lies well within the earth mass, the thrust on it must be taken as acting parallel to the top slope, and its amount will be the same as that given by the Rankine theory.

Although it is highly desirable to have more precise experiments on large models in order to draw sure conclusions, yet, as far as the experiments go—those which have been analyzed and discussed in this paper—the following conclusions may be stated:

1.—When wall friction and cohesion are included, the sliding-wedge theory is a reliable one, when the filling is a loosely aggregated granular material, for any height of wall.

2.—For experimental walls, from 6 to 10 ft. high, and greater, backed by sand or any granular material possessing little cohesion, the influence of

cohesion can be neglected in the analysis. Hence, further experiments should be made only on walls at least 6 ft., and preferably 10 ft., high.

3.—The many experiments that have been made on retaining boards less than 1 ft. high, have been analyzed by their authors on the supposition that cohesion could be neglected. This hypothesis is so far from the truth that the deductions are very misleading.

4.—As it is difficult to ascertain accurately the coefficient of cohesion, and as it varies with the amount of moisture in the material, small models should be discarded altogether in future experiments, and attention should be confined to large ones. Such walls should be made as light, and with as wide a base, as possible. A triangular frame of wood on an unyielding foundation seems to meet the conditions for precise measurements.

5.—The sliding-wedge theory, omitting cohesion but including wall friction, is a good practical one for the design of retaining walls backed by fresh earth, when a proper factor of safety is used.

As the subject of pressures on the roof and sides of a tunnel lining has received much attention of late, the writer has concluded to extend this paper, so as to give a development of a theory, based on the grain-bin theory of Janssen, but modified to include the cohesive or shearing resistances of the earth in addition to the frictional resistances.

[Fig. 16](#) is a vertical transverse section of a tunnel, $ABCD$, and the earth, $DCFE$, extending over it h feet. If this tunnel has been driven by the use of a shield or poling boards, the ground will tend to settle over it, and part of the weight of $DCFE$ will be sustained by cohesion and friction (resulting from the lateral thrust) exerted along the sides, vertically upward. The earth will probably arch itself, or form a series of domes superposed one upon the other, but the external forces acting on such domes will be the same as those acting on a corresponding horizontal lamina, and the theory,

given in full in the Appendix, begins with the considerations pertaining to the equilibrium of such a lamina.

If there was no settlement of the earth, $DCFE$, in relation to DC , then the vertical pressure per square foot on DC would be $w h$ (w being the weight of a cubic foot of the earth in pounds), but, as most of the weight of $DCFE$ is carried by the sides, in case of sufficient settlement, the vertical unit pressure, V , on DC , will be much less than $w h$. Also, the lateral unit pressure, L , at the level, DC , will be much less where settlement occurs. From the equations for V and L , given in the Appendix, the diagrams, [Figs. 17](#) and [18](#), have been constructed.

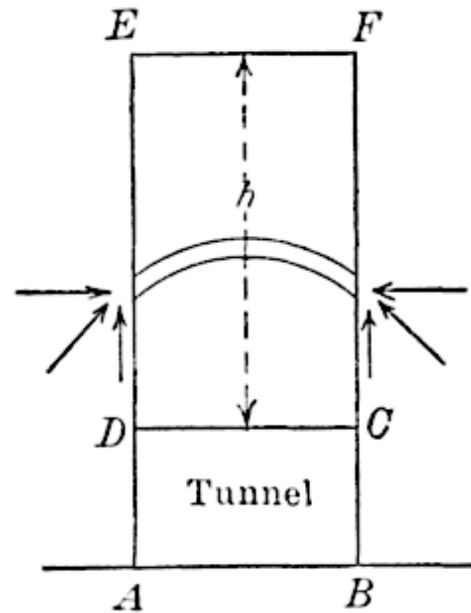


FIG. 16.

In both diagrams, the weight of the earth was taken at $w = 90$ lb. per cu. ft., and the cohesion of the earth at $c = 100$ lb. per sq. ft. In [Fig. 17](#), $\phi = 45$ and the curves for V and L were laid off for a width of tunnel, b , of 15 ft. and also for 30 ft. In [Fig. 18](#), $\phi = 30^\circ$, and curves are given for V and L , also for $b = 15$ ft. and 30 ft. for various heights, h .

It will be perceived, in both figures, that when certain heights are attained, both V and L cease to increase perceptibly, so that such values may be taken as corresponding to h indefinitely large.

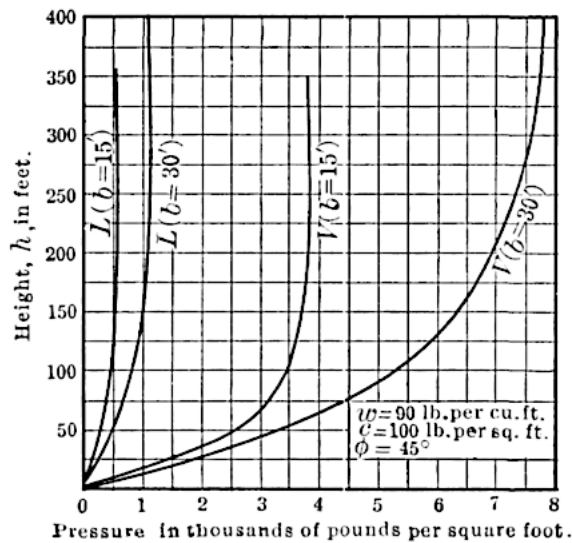


FIG. 17.

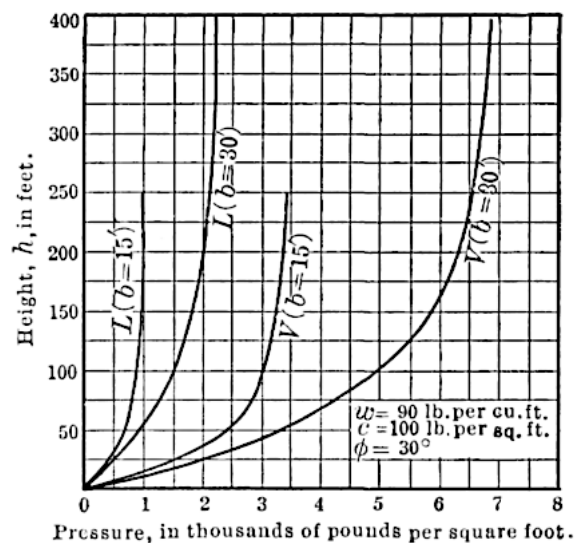


FIG. 18.

A simple way of deriving these extreme values is given in the Appendix. The values of ϕ , b and w have been taken here the same as those used by Mr. Meem, in framing his table of pressures,¹³ which may be supposed to embody, in part, practical experience. The results found from Figs. 17 and 18 by the writer, for a depth of covering of several hundred feet, are uniformly much larger than those given by Mr. Meem. Are they too large for safety? In answering this question, it must be remembered that, of the weight of earth directly over the tunnel, all has been transferred to the sides that it was possible to transfer, for the coefficients of friction and cohesion given. We know scarcely anything of the cohesion coefficients, so that the value assumed, $c = 100$ lb. per sq. ft., may not be near the truth. Certainly it must appear plain from this discussion that the values of c and ϕ must be better known, for all kinds of earth, before reliable results can be attained. The results are submitted for discussion, in the hope that engineers will give their experience relative to the pressures realized in the timbering of tunnels, particularly through sand or earth not thoroughly consolidated.

The value of V , in Figs. 17 and 18, is the average vertical unit pressure at the top of the tunnel. Experiments on grain bins lead to the inference that

the pressure at the middle of the roof is greater than that at the sides, but no law of variation can be stated.

The lateral unit pressure on the vertical sides of the tunnel lining at the top is given by the equation for L , or by the corresponding diagram. The variation in this lateral pressure over the sides of the tunnel cannot be easily formulated, as so much of the weight of the earth, directly over the tunnel, has been transferred by a kind of arch action to the sides. Experience would better speak here.

Table 5 gives the values of V and L for $h = 40$ ft. The figures in Columns M are taken from Mr. Meem's table, previously referred to; those for Columns C are from the diagrams, Figs. 17 and 18.

In quoting Mr. Meem's figures, the writer must not be understood as endorsing in any way his theory; but the results are of interest as embodying the conclusions of a practical engineer of large experience.

TABLE 5.

b , in feet.	ϕ .	V , IN POUNDS PER SQUARE FOOT.		L , IN POUNDS PER SQUARE FOOT.	
		M .	C .	M .	C .
15	45°	1 485	2 300	405	300
15	30°	1 035	2 100	540	600
30	45°	3 240	2 800	450	400
30	30°	2 325	2 600	450	750

If the height, h , of earth covering is 200 or 300 ft., the values given by Figs. 17 and 18 are much larger than those given in Columns M , which presumably represent Mr. Meem's pressures for any height greater than 40 ft.

In saturated earth, it has been customary, perhaps, to regard the earth as if it were gravel composed of solid spheres, like marbles, so that the water

has free access in any direction. Thus, in the case of a retaining wall backed by such material, the water has full access practically to every part of the wall, and the wall is subjected to the full water pressure corresponding to its depth. It is likewise subjected to a thrust from the earth, corresponding to c and ϕ , for the saturated material, but with a weight per cubic foot equal to that of the earth in air less the buoyant effect of the water. Thus, if a cubic foot of the porous earth, in air weighed 90 lb., and if the voids were 40%, then 1 cu. ft. of earth contains 0.6 cu. ft. of solids and the buoyant effect of the water is the weight of an equal volume of water or $0.6 \times 62.5 = 37.5$ lb. Hence, the weight per cubic foot of earth in water is $90 - 37.5 = 52.5$ lb.

Similarly, for the pressures, V and L , at the top of a tunnel, $w = 90$ must be replaced by 52.5, and c and ϕ must be found for the saturated material and these values substituted in [Equations \(5\) and \(6\)](#) of the Appendix. To these pressures must be added the corresponding water pressures for the full height of water, supposing it to have free communication everywhere, as in the case of the gravel filling. However, with sand, or earth with much fine material, the pores are more or less clogged up and there is perhaps intimate contact of a part of the earth with the roof of the tunnel, so that the water cannot get under it to produce a lifting effect, and if such intimate contact is found along any horizontal or vertical section, of the earth on either side of the section, it is plain that the buoyant effort of the water on a cubic foot of material will be much diminished.

Mr. Meem deserves great credit, not only for calling attention to this, but especially for performing certain experiments to prove it.¹⁴ The experiments were on sand, and only on a small scale, but the practical conclusion drawn from them is that the water pressure transmitted through sand having 40% voids is diminished about 40% in intensity. This occurs for a depth of only a few inches of sand, and presumably the diminution

would be greater for sand several feet in depth. Of course, before definite values can be stated, experiments on a large scale should be made on every kind of material usually met; but, as a numerical illustration of the application, for the diminution mentioned—which is assumed to extend through the mass—it is seen that, in the examples of the retaining wall and also the tunnel, the weight per cubic foot of the earth in water must now be taken at $90 - (0.4 \times 0.6 \times 62.5) = 75$ lb. per cu. ft.

This value replaces the w in [Equations \(5\) and \(6\)](#) from which the V and L for the top of the tunnel are found. To these values, add $0.4 \times 62.5 h_0$, for the water pressure, where the surface of the water extends a height, h_0 , above the top of the tunnel. Similarly, in the case of the retaining wall, add 0.4 of the full water thrust on the wall to that given by the earth, weighing only 75 lb. per cu. ft.

As a numerical illustration, take $b = 15$ ft., $\phi = 45^\circ$, w in air = 90 lb., $h = 40$ ft., $h_0 = 60$ ft.; but we must now replace w by the weight in water, 75 lb., as found above. The values of V and L are now found, by [Equations \(5\) and \(6\)](#) ([Appendix](#)), to be 1 917 and 246 lb. per sq. ft., respectively, for the saturated earth alone. To these values add $0.4 \times 62.5 \times 60 = 1\,500$ lb. per sq. ft., water pressure, giving a total of 3 417 and 1 746 lb. per sq. ft., respectively, for the vertical and horizontal unit pressures at the top of the tunnel lining.

In connection with this subject of underground pressures, it may not be inappropriate to make some concluding remarks on the maximum vertical pressures to which culverts may be subjected.

Let [Fig. 16](#) now represent a longitudinal vertical section along the axis of a road embankment, built over an arch culvert or box-drain, $ABCD$, the line, CD , passing through the summit of the arch or the top of the covering stone of the box-drain, and the lines, AD and BC , coinciding in part with the exterior sides of the abutments.

There is a horizontal thrust of the earth on the medial plane, $CDEF$, acting at right angles to the plane of the paper, which tends to distribute the weight of the central portion partly toward the sides; but, ignoring this, it is seen that, if the earth everywhere settles uniformly, the maximum pressure per square unit at the top of the culvert is $w h$, and the total vertical pressure on the culvert is the weight of the earth vertically above it.

If, however, the earth outside the abutment walls settles more than the walls (a case which may occur), then part of its weight, and that of the earth vertically above it, will be transferred, through friction and cohesion, along the planes, AE and BF , to the culvert, and thus the vertical pressure on the top of the culvert will be greater than in the first supposed case; but, if the reverse obtains, or if the culvert settles more than the earth outside the lines, AD and BC , or if the arch or covering stone descends in the middle relatively to the abutments, then part of the weight of the earth vertically over the culvert is transferred to the sides. For a comparatively rigid arch, the settlement is perhaps not enough to warrant us in making the maximum unit pressure less than $w h$. Exactly what settlement would warrant the use of the theory set forth in the Appendix it is impossible to say. If the unit pressure is taken as $w h$, we can rest assured that in most cases the real pressure is materially less.

1 The writer refers to his “Retaining Walls,” Van Nostrand’s Science Series, No. 3, for the demonstrations pertaining to the above constructions, and to the derivations of formulas.

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2 A full discussion may be found in the writer’s “Retaining Walls.”

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3 The experiments pertaining to [Figs. 7, 8, and 9](#) are due to Curie. See Curie’s “Poussée des Terres” and “Trois Notes,” Gauthier-Villars, Paris. They are of especial interest in that they were undertaken to attempt to overthrow the theory advocated above.

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4 All the experiments of Leygue referred to in what follows may be found in *Annales des Ponts et Chaussées*, November, 1885.

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5 We can suppose, here, the horizontal force to be the pull of a cord extending horizontally from the box and passing over a fixed pulley, and that at the free end of the cord a weight is applied. The friction of the pulley and carriage wheels could be found experimentally and allowed for, so that some fraction of this weight would equal P .

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- 6 This method is an extension of that given by Professor H. T. Eddy in his treatment of earth thrust, in “Researches in Graphical Statics.” [Return to text](#)
- 7 To attain the greatest accuracy, in constructions like that shown in [Fig. 11](#), the scale should be as large as possible; the arcs of circles, especially, must be drawn with a large radius, and the points, s_1 , s_2 , etc., determined with care. The angle, $H A f$, can be computed and laid off by aid of a table of chords. The construction in this figure corresponds to a vertical height of $A B = 0.656$ ft., $k = 1$, $e = 80$. The value of the component, E_1 , perpendicular to $A B$, is now to be found, by drawing lines from n and c_4 , perpendicular and parallel to $A B$, to intersection, and measuring the component to scale. For $k = 2$, it is found that $A b_1$ is the plane of rupture. The line, $m c_1$, through the new c_1 (c'_1), representing the thrust, is very small; but it can be easily magnified by laying off the polygon, $g 1 m c'_1$, to a scale two or three times as large, and thus the thrust can be found as accurately as before. [Return to text](#)
- 8 *Minutes of Proceedings*, Inst. C. E., Vol. LXV. p. 140: reprinted in Van Nostrand’s Science Series. [Return to text](#)
- 9 In reference to this equation, see Appendix. [Return to text](#)
- 10 “The Bracing of Trenches and Tunnels, With Practical Formulas for Earth Pressures.” *Transactions*, Am. Soc. C. E., Vol. LX. p. 1. A number of important facts brought out in this paper are of vital importance to constructors. [Return to text](#)
- 11 “Mechanics of Materials,” Tenth Edition, p. 381. [Return to text](#)
- 12 Perhaps this may be accounted for by supposing cohesion between the blocks at rest, which is destroyed by the motion, when only friction acts. [Return to text](#)
- 13 *Transactions*, Am. Soc. C. E., Vol. LXX, p. 387. [Return to text](#)
- 14 *Transactions*, Am. Soc. C. E., Vol. LXX, pp. 365–368. [Return to text](#)

A P P E N D I X.

In the experiments of Jamieson and Pleissner on the pressures in deep grain bins¹⁵, the ratio, k , of the lateral unit pressure, L , on a vertical plane to the vertical unit pressure, V , on a horizontal plane, was found by Pleissner to vary from 0.3 to 0.5 and by Jamieson to equal 0.6, for wheat in wooden bins of various sizes.

This ratio, $k = \frac{L}{V}$, increases somewhat with the depth of the grain, but the increase is slight after a depth of from $2\frac{1}{2}$ to 3 times the width or diameter of the bin is reached.

It is recognized that the proper value of k , for a particular case, can only be determined properly by experiment, but it is interesting to note that, by the theory of earth pressure of an unlimited granular mass, level at the top, the ratio of the lateral to the vertical unit pressure, at any point in the mass, is, $k' = \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$, and that this varies from 0.361 to 0.271, as ϕ , the angle of repose, varies from 28° to 35° , the values of ϕ for wheat, given in some of the experiments. Further, by reference to Jamieson's experiments on a model bin of smooth steel, 1 ft. in diameter,¹⁶ filled with sand, for which $\phi = 34^\circ$, $k' = \tan^2 \left(45^\circ - \frac{\phi}{2} \right) = 0.283$, we find the

experimental value of k to equal k' exactly for a height of sand of 2.5 ft., the value at 6 ft. and upward being 0.33.

The theory of bin pressure is utterly different from the ordinary theory of earth pressure in an unlimited granular mass; but it is seen that the latter may be of some use in furnishing a value of k when experimental values are lacking, as in the case of various kinds of earth, both granular and more or less consolidated.

An equation for L , for an unlimited mass of earth, level at the top and having a coefficient of cohesion, c , has been given by Scheffler,¹⁷ and is as follows:

$$L = w h \tan^2 \left(45^\circ - \frac{\phi}{2} \right) - \frac{c \cos \phi}{\cos^2 \left(45^\circ - \frac{\phi}{2} \right)}. \quad (1)$$

Here w = weight per cubic unit of earth,

h = depth below the level-topped, unlimited mass,

L = unit lateral pressure on a vertical plan at the depth, h .

This reduces to the usual formula when $c = 0$.

It is seen from [Fig. 19](#), if we lay off at the depth, h , $BC = w h \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$, that the horizontal ordinates of the triangle, ABC , measure the values of the first term of the right member of [Equation \(1\)](#). The second term, $\frac{c \cos \phi}{\cos^2 \left(45^\circ - \frac{\phi}{2} \right)}$, is constant, and is

represented by the horizontal ordinates of the rectangle, $ABGF$. Thus the value of L at the depth, h , is represented by $BC - BG = CG$.

At D , $L = DE - DE = 0$; but, above D , the equation is inapplicable, for negative values of L , corresponding to tension along AD , are inadmissible; hence, above D , we must write $L = 0$.¹⁸

In Equation (1), for $h = h' = AD$, $L = 0$, therefore

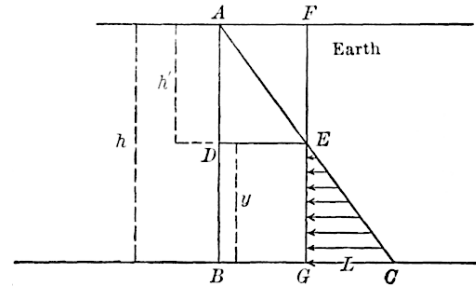


FIG. 19.

$$0 = w h' \tan^2 \left(45^\circ - \frac{\phi}{2} \right) - \frac{c \cos \phi}{\cos^2 \left(45^\circ - \frac{\phi}{2} \right)}. \quad (2)$$

Solving this for h' ,

$$h' = \frac{c}{w} \frac{2 \cos \phi}{1 - \sin \phi}. \quad (3)$$

For given values of c , w , and ϕ , having computed h' , we have, on subtracting Equation (2) from Equation (1),

$$L = w (h - h') \tan^2 \left(45^\circ - \frac{\phi}{2} \right) \quad (4)$$

which is true only for $h = h'$ or $h > h'$. When $h < h'$, $L = 0$.

If we put $y = (h - h')$, the equation reduces to the ordinary form, and thus the center of pressure of the thrust on AB (or DB) acts at $\frac{1}{3} y$ above B , and its amount is $\frac{1}{2} w y^2 \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$.

Call V , the vertical unit pressure at the depth h ,
 V' , the vertical unit pressure at the depth h' ,
therefore $V = w h = w(y + h')$,

$$V' = w \, h';$$

and, putting $k = \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$, we have, from Equation (4),

$$L = w(h - h') \tan^2 \left(45^\circ - \frac{\phi}{2} \right) = (V - V')k.$$

Next consider the case of a tunnel of width, b , and length, l , which has been driven by shield or by use of timbering, so that an appreciable settlement of the roof occurs; then the weight of the earth vertically over the tunnel is partly carried by the adjacent walls of earth, by friction and cohesion, and it would seem that such walls can be supposed to take the place of vertical grain bin walls, and that the theory of bin pressures corresponding may be made to apply. The theory that will be developed, which includes the influence of cohesion, is simply a modification of that used in developing Janssen's formula, as given by Mr. Ketchum,¹⁹ and, for a ready comparison of results, his notation will be used.

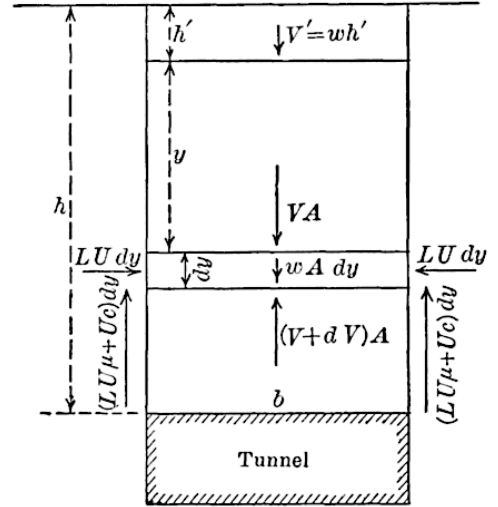


FIG. 20.

In Fig. 20, let h be the distance from the roof of the tunnel to the surface of the earth. Compute h' from Equation (3) and lay it off from the top down.

Let V = vertical unit pressure at depth, $y + h'$;

L = lateral unit pressure at depth, $y + h'$;

w = weight of earth per cubic unit;

ϕ = angle of repose of earth;

$\mu = \tan \phi$ = coefficient of friction of earth on earth;

$A = b l$ = area of horizontal section of tunnel;

$U = 2(b + l)$ = perimeter of section; and

c = cohesive force of earth per square unit.

The foregoing equation, $L = (V - V') k$, will be used as a semi-empirical formula to express the relation between L and V , but now V is no longer equal to $w (y + h')$ but at present is unknown. As before, $V' = w h'$.

Properly, k should be determined by experiments, but, from lack of such experiments, it will be computed from the formula above,

$$k = \tan^2 \left(45^\circ - \frac{\phi}{2} \right).$$

Consider now the conditions of equilibrium of a horizontal slice of earth of depth, dy , the weight of which is $A w dy$.

The top surface, at the depth, $y + h'$, is acted on by the force, VA , acting downward, and the bottom surface by the force, $(V + dV) A$, acting upward. The total lateral force acting on the area, $U dy$, is $L U dy$ and this causes a frictional force of $L U \mu dy$, acting upward. The cohesive or shearing resistance on the area, $U dy$, acts upward, and its amount is $U dy c$. Placing the sum of the vertical forces acting on the slice equal to zero.

$$V A + A w dy - (V + dV) A - (L U \mu + U c) dy = 0.$$

In reality, an arch or dome of the earth should be considered in place of the horizontal stratum, but the result is the same, because the same vertical forces act in either case. Simplifying the above equation, and dividing by A ,

$$dV = \left[w - (L \mu + c) \frac{U}{A} \right] dy.$$

Putting $R = \frac{A}{U}$, and $L = (V - V') k$,

$$\begin{aligned} \frac{dV}{dy} &= w - [(V - V') k \mu + c] \frac{1}{R} \\ &= w - \frac{k \mu V}{R} - \frac{c - V' k \mu}{R} = w - n V - q, \end{aligned}$$

on placing, $n = \frac{k \mu}{R}$, $q = \frac{c - V' k \mu}{R}$.

It follows that, $\frac{-n d V}{w - n V - q} = -n dy$
therefore $\log(w - n V - q) = -n y + C$.

When $y = 0$, $V = V'$, therefore

$$\begin{aligned}\log(w - n V' - q) &= C, \\ \log \frac{w - n V - q}{w - n V' - q} &= -n y, \\ \text{and } \frac{w - n V - q}{w - n V' - q} &= e^{-n y},\end{aligned}$$

where $e = 2.71828 \dots$, the Napierian base.

Solving for V ,

$$V = \frac{w - q}{n} [1 - e^{-n y}] + V' e^{-n y}.$$

Substituting the values of n and q , we have,

$$V = \frac{1}{k \mu} [R w - (c - V' k \mu)] \left(1 - e^{-\frac{k \mu}{R} y}\right) + V' e^{-\frac{k \mu}{R} y} \quad (5)$$

in which it must be remembered that,

$$v' = w h'; h' = \frac{2 c \cos \phi}{w(1 - \sin \phi)}.$$

The lateral thrust is now given by,

$$L = (V - v') k. \quad (6)$$

To get the pressures, V and L , at the top of the tunnel, replace y by $(h - h')$.

The weight of the upper stratum, of depth h' , is in part sustained by the cohesion of the sides, but as h' is generally small, this cohesive force can be

neglected, as was done above.

Equations (5) and (6) reduce to the ordinary bin formulas of Janssen, when $\mu = \mu'$, $c = 0$, and therefore $V' = 0$, $h' = 0$. The modification due to these terms is generally small, unless c is very large.

For large values of y , $e^{-\frac{k\mu}{R}y}$ is small, and as y increases indefinitely, V approaches as a limit the value

$$\frac{1}{k\mu} [R w - (c - V' k \mu)].$$

This expression may be derived independently, and is of practical value when a very high surcharge is considered.

Referring to Fig. 20, it is evident that the maximum limit of V would be realized if the weight of any horizontal lamina is entirely held up by the friction and cohesion of the sides; for then, for all lower slices, V and L remain the same.

$$\begin{aligned} \text{Therefore } A w dy &= (L U \mu) + U c) dy \\ A w &= (V - V') k U \mu + U c, \\ \text{therefore } V &= \frac{1}{k\mu} [R w - (c - V' k \mu)], \end{aligned}$$

as given above.

As seen, such a state is not exactly realized, but is practically true for great depths.

For a long tunnel, the perimeter of the section, U , can be taken as $2l$, whence,

$$R = \frac{b l}{2 l} = \frac{b}{2}.$$

This value was used in all the computations.

As a numerical illustration of the use of [Equations \(5\) and \(6\)](#), suppose a tunnel, $b = 15$ ft. wide, and, for the earth covering let $\phi = 45^\circ$; therefore $\mu = \tan \phi = 1$, $w = 90$ lb. per cu. ft., and $c = 100$ lb. per sq. ft.

$$\begin{aligned} \text{We deduce } k &= \tan^2(45^\circ - 22\frac{1}{2}^\circ), & R &= \frac{b}{2} = 7.5, \\ h' &= \frac{2 c \cos \phi}{w (1 - \sin \phi)} = 5.4 \text{ ft.}, & \text{therefore } V' &= w h' = 486 \text{ lb.}, \\ (c - V' k \mu) &= 100 - 83 = 17. \end{aligned}$$

[Equations \(5\) and \(6\)](#) readily reduce now to

$$\begin{aligned} V &= 3830 (I - e^{-0.023 y}) + 486 e^{-0.023 y}, \\ L &= (V - 486) \times 0.172. \end{aligned}$$

These formulas give the vertical and horizontal unit pressures at the top of the tunnel when,

$$y = h - h'.$$

In computing the values of V and L for various depths of earth covering, h , a short table of hyperbolic logarithms is a convenience. The curves given by the equations above are shown on [Fig. 17](#).

An additional note with respect to [Equation \(1\)](#) may not be inappropriate. Scheffler, in deriving this equation, considered the conditions of equilibrium of an infinitesimal wedge of earth at the depth, h . It was found that the horizontal pressure at the depth, h' , given by [Equation \(3\)](#), was zero, and it was assumed by the writer that there was no pressure on a vertical plane for a less depth. Thus, in [Fig. 21](#), there is no horizontal pressure on the plane, DE , where $DE = IC = h'$; consequently, the weight of the wedge, EDB , is supported entirely by the normal reaction of the plane, DB , with the cohesion and friction acting along it.²⁰ To deduce Q , the total earth pressure on the vertical plane, AC , it is then admissible to treat the prism, $ADEC$, as the prism of rupture, the surface of rupture consisting of the plane, AD , making the angle α with the vertical and the

plane, DE . Therefore W , the weight of the prism, $ADEC$, is in equilibrium with Q , N , and $N f + K$, where N is the normal reaction of AD , $f = \tan \phi$, and $K =$ the total cohesion on $AD = c \cdot AD$.

On balancing components parallel and perpendicular to the plane, and then following familiar methods, it can be shown that the true value of Q corresponds to $\alpha = \frac{90 - \phi}{2}$, and that this value is,

$$Q = \frac{w}{2} (h^2 - h'^2) \tan^2 \left(\frac{\pi}{4} - \frac{\phi}{2} \right) = \frac{c (h - h') \cos \phi}{\cos^2 \left(\frac{\pi}{4} - \frac{\phi}{2} \right)}.$$

The derivative of this, with respect to h , gives the intensity, L , at the depth, h , exactly the same as Equation (1), and the subsequent deductions hold. Thus the fundamental Equation (1), according to the interpretation given, is seen to correspond to a prism of rupture, $ADEC$, which is a little nearer the true one, having a curved surface of rupture, than the wedge, ABC .

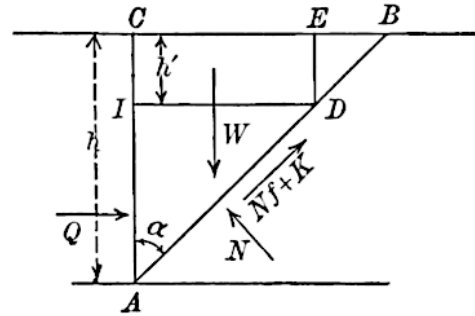


FIG. 21.

The above refers to the pressure on a vertical plane of a mass of level-topped earth of indefinite extent; but suppose that AC is the back of a retaining wall, and that a slight movement downward of the prism of rupture is imminent; then, if the earth along the plane, DE , can exert sufficient tension, the mass, $ADEC$, in descending, may drag down the wedge, DEB , with it, so that the full friction and cohesion along DB will be added to that along AD . In other words, the prism of rupture must now be taken as the wedge, ABC ; hence, the value of Q corresponding is given by the equation above, on making $h' = 0$, as this introduces, in the first equations for equilibrium, the fact that the prism of rupture is now the wedge, ABC .

It is only one step farther to find the greatest height at which the vertical face of an open trench will stand for given coefficients, c and f . On making $Q = 0$ in the equation for Q above when $h' = 0$, we find, after reduction,

$$h = \frac{4 c \cos \phi}{w (1 - \sin \phi)},$$

a value which has been quoted elsewhere in this paper. It is double the value for h' given by [Equation \(3\)](#). The reason for this, though, is now evident; for the last equation follows as a consequence of assuming that the full cohesive and frictional resistances along DB were exerted; whereas [Equation \(1\)](#) ignores them.

15 Given in detail in “The Design of Walls, Bins and Grain Elevators,” by Milo S. Ketchum, M. Am. Soc. C. E. [Return to text](#)

16 Ketchum’s “Walls, Bins, and Grain Elevators,” Fig. 171. [Return to text](#)

17 “*Traité de Stabilité des Constructions*,” p. 292; see also Remark at end of Appendix. [Return to text](#)

18 Scheffler has not noted this fact, and consequently some of his deductions are open to objections. His theory, involving cohesion, is the only one the writer has seen. [Return to text](#)

19 “The Design of Walls, Bins and Grain Elevators,” Chapter XVI. [Return to text](#)

20 It may be well to remark here, that for cohesive earth, it has been proved, both theoretically and experimentally, that the surface of rupture is curved, and not a plane, as the theory assumes. However, assuming it to be a plane, and considering successive wedges of rupture of different heights, the bases of which lie on the same plane, it can be easily shown that certain of the upper wedges can be sustained by cohesion alone, and that the coefficient of cohesion required for stability varies from 0 at the surface to its maximum value at a certain depth, h_1 . Below this depth, friction in addition to cohesion is exerted, and stability is assured if we suppose the friction coefficient to increase from 0 at $h = h_1$ to its maximum value, $\tan \phi$, at some depth, $h = h'$. Below this depth, on the plane of rupture, the maximum values of both coefficients are exerted. Now, the ordinary wedge theory assumes, for simplicity, that these coefficients are constant all along the plane of rupture, which may be true at the instant of rupture, but not for a stable mass. It is possible, too, that rupture may be progressive, starting at the bottom. [Return to text](#)

DISCUSSION.

J. R. WORCESTER, M. AM. SOC. C. E. (by letter).— In reading Professor Cain's admirable paper relating to experiments on retaining walls, the writer has looked in vain for a word of caution as to the effect which time plays in modifying the condition of equilibrium within a mass of earth. The author evidently considers it necessary to allow (by using a factor of safety) for a possible lessening of the angle of friction on account of a change in the amount of moisture, and possible vibrations, but states that in a great majority of cases the greatest thrust will occur where the earth has been recently deposited. It would appear that he neglects a possibility, if not a probability, of a readjustment of the earth particles through the influence of time, by which the angle of friction is lessened if not wholly cancelled.

The theory that cohesion in the earth and frictional resistance on the back of the retaining wall account for the experimental results seems indisputable, but such experiments must needs be carried through in a reasonable time, and, in that respect, at least, must needs differ from actual constructions which are intended to be permanent.

It is well known that unbraced excavations can often be carried vertically to considerable depths in safety, but that not infrequently— as many have learned to their sorrow— such unbraced banks have subsequently caved in. The slides in the Culebra cut may be mentioned as a

similar illustration. To be sure, the delayed motion of the earth (or rock) may be attributed to the effect of moisture, but that does not invalidate the argument, as one always has to reckon with water.

It is also a matter of common knowledge that, in braced excavations, the pressure on the sheeting and bracing frequently increases in time to an extent enormously exceeding the original pressure. In many instances this has caused a failure long delayed.

Another instance of the effect of time is found in many retaining walls, in which a very slow motion has occurred, though the walls appeared to be entirely stable when first built.

A frictional or tangential force along the back of the wall may tend to prevent motion, but it is difficult to conceive of this tangential force being perpetually present on the back of a stationary wall after the back-filling has become settled and consolidated. In the interior of the mass of earth at rest the author admits that the reaction between particles is not along inclined lines, but rather that the lines of pressure are vertical and horizontal. The vertical lines of force are, of course, caused by gravity, and the horizontal lines by the tendency of particles to wedge in between those below and to spread them.

If this conception of the forces within a mass of earth is reasonable, it would seem as if it might also be extended to the pressure against an immovable vertical wall. One must then consider whether it is possible for a horizontal pressure to cause the wall to move, without changing the conditions and introducing the inclined stresses. It would seem within the bounds of possibility that a very minute motion might be produced, and that this would be followed by a readjustment of stresses in the earth by which the forces would gradually resume the horizontal direction.

The nature of the soil undoubtedly has much to do with this question. In some kinds of clay there appears to be a sort of viscosity, such as is

frequently seen in pitch and other materials, or a tendency toward a slow flowing. No amount of pressure would cause a sudden motion, but time will effect a motion under a slight pressure or even the force of gravity alone. It appears that this condition is produced by the very minute particles, each moving individually into a position in which the surrounding forces balance. If one cuts a vertical face in such material one cannot force the exposed particles out of their position by crowding them from behind, but each in its turn will feel the pressure unbalanced and will slowly move out. This may not be true to the same extent with granular materials of large diameter, but a familiar instance is seen in fine wet sand. If a small excavation is dug in a wet beach sand, the banks will stand vertically at first, but, by watching closely, one may see the particles, beginning at the foot of the bank where there is most water, gradually moving out, overcoming the force of cohesion, and ever tending to seek a condition with a level surface. It seems quite likely that a similar tendency would exist in almost all soils, to greater or less degree, though perhaps it might be safely neglected in a mass of hard, irregularly-shaped fragments of stone which could interlock.

The point which the writer wishes to make is that a word of caution should accompany this argument for the frictional and cohesive forces; that they cannot always be relied on; and that sometimes the Rankine theory may be better than the wedge theory in designing, even though it does not seem to fit the experimental results.

Another warning may not be amiss, in considering the safe thickness to allow for retaining walls, and that is the effect of frost, where the surface of the ground is level and likely to retain moisture. The swelling force of freezing, under these circumstances, may be more than sufficient to overcome the beneficial effects of both cohesion and friction. Presumably this must be provided for in the “factor of safety,” and is in itself a justification for a very appreciable factor.

It may be well to emphasize the fact that a large part of the author's assumed factor of safety seems to be absorbed in keeping the resultant within the middle third of the base. The proportions between width of wall and height, determined on [pages 433](#) and 434, are such as to keep this resultant just within the base. If, with these same proportions of wall, the factor were assumed so that the resultant were within the middle third, it would be found to be nearer $1\frac{1}{4}$ than 3. The author's statement on [page 435](#), that he "does not advocate the middle-third limit method in design," is not wholly clear, but the implication is that the resultant should be well within this limit. In this case, it seems as if the factor of safety would be wholly absorbed in thus locating the resultant, and would leave nothing for other elements of uncertainty.

J. C. MEEM, M. AM. SOC. C. E. (by letter).—In the writer's judgment the author has gone a step forward in developing the relation of cohesion and friction to walls and tunnels, but he has not given sufficient value to the larger consideration of what may be called "cohesive friction" induced by the lateral pressure of earth against retaining walls and other faces. This will be noted later in the discussion.

The author states that "from all that precedes, it is seen that the results of experiments on small models in the past have proved to be very misleading, and that experiments on large models are desirable, and can alone give confidence."[21](#) To a certain extent this is fully in accordance with the writer's views, as noted in his papers on earth pressures,[22](#) and he feels justified in once more calling attention to the fact that, in his judgment, the only experiments which can definitely establish the value of earth pressures against walls and sheeted faces should be made on a large scale and against independently laid and independently braced horizontal sheeting. If these experiments are made in a homogeneous material, such as dry sand of a known angle of repose, it is believed that it will be conclusively and definitely shown that the pressures at a point above the middle plane will be

greater than below it, and, further, that it will be proved that the pressure near the bottom, for example, of a 20-ft. trench, will not be perceptibly greater than that against a brace at the same distance from the bottom of a 40- or a 60-ft. trench.

The writer agrees with the author that the theory of a sliding wedge is the best practical one for retaining walls, and if the face of the wedge be keyed tight, as stated by the author, or through the compacting of the material into a more solid mass, it will be seen that, with no break, the resultant pressures against the wall or face are virtually those, not only of a sliding, but of a solid wedge tending to slide on the plane of repose, the mass being compactly held together by cohesion induced by lateral pressure. Although the general solidity of this mass is dependent on the stability of the wall or bracing, the pressures caused by the tendency of this solid wedge to slide are not affected materially by slight changes in form due to gradual settlement, which, in turn, may be caused by the normal yielding of the wall or shrinkage of the bracing, or by any small losses of material, as long as any of those or their sum is not sufficient to cause partial or final collapse. Assuming then that this theory of the solid wedge tending to slide is the true one, it is difficult to reconcile it with the results of experiments on revolving boards or revolving walls, except in so far as they relate to the more accurate determination of the coefficients of friction and cohesion.

In connection with the experiments of Leygue, the writer wishes to refer to the dry sand and wheat arching experiments described in the final discussion of his paper, "The Bracing of Trenches and Tunnels, with Practical Formulas for Earth Pressures,"²³ and especially to the experiment made with dry sand in a 2-in. pipe, described in his paper on "Pressure, Resistance, and Stability of Earth."²⁴ For clearer reference, the latter experiment is again described:

A 2-in. pipe, 18 in. long, was filled with dry sand for a depth of 12 in., and a thin piece of tissue paper was pasted across the bottom. Then, with a wooden piston bearing on the sand, the latter would support the blow of a sledge hammer or the weight of a man without breaking the tissue paper.

All these experiments seem to prove very clearly that pressure is distributed laterally, at least through perfectly dry material, and, with certain values known, it should not be difficult to compute the value of this thrust. For example, if, in the pipe experiment, the exact coefficient of friction of the granular particles against the inside of the pipe is determined to be 0.4, then it would take a pressure equal to the complement of this, or 0.6 of the weight, to hold the granular particles against the vertical side. Now, if the exact depth of sand is determined at which no further pressure (by reason of adding to this depth) is transmitted to the bottom, then, by finding the difference or loss of pressure at the bottom, it should be a simple matter to compute the amount of thrust developing sufficient friction over the given area to sustain the additional weight. This is what the writer, for want of a better term, has called “cohesive friction” induced by lateral pressure. While such experiments on a small scale give definite results with very dry sand or wheat, they will not be conclusive unless made on a very large scale in cases where dampness adds materially to the normal cohesion of small masses.

As to the experiments on revolving boards or walls referred to by the author, it would appear that, as noted, they eliminate consideration of the effect of pressure as increasing the values of the coefficient of friction and cohesion.

The writer believes that it can be shown by [Fig. 22](#) that the lateral pressure against sheeted faces or retaining walls is greater toward the top. [Fig. 22](#) is adapted from Fig. 5 in the writer’s paper, “Pressure, Resistance, and Stability of Earth.”

It is supposed that a retaining wall has first been built along the face, $A B$, and that it has been back-filled with ordinary dry sand to the curved line $F C$; this is then covered with a heavy canvass or tarpaulin, and Rods 1, 2, 3, etc., are run through it in sufficient numbers to hold it in position when the area, $A F C D$, has been back-filled with dry sand. The tarpaulin is then keyed up tight by the rods bearing on washers of large area on the top plane, $A D$. If the sand in the area, $F E C$, be then removed, the observer is asked to note whether, in his judgment, the resulting pressure is greater at P or at P' . It must undoubtedly be true that if this experiment establishes the writer's claim in the case of normally dry sand, it will tend to do so much more readily in the case of mixed sands and earths as ordinarily found in trenches. If this fact of the greater pressure toward the top be proved conclusively, as the writer believes it undoubtedly will be in time, it may account for some of the failures of retaining walls.

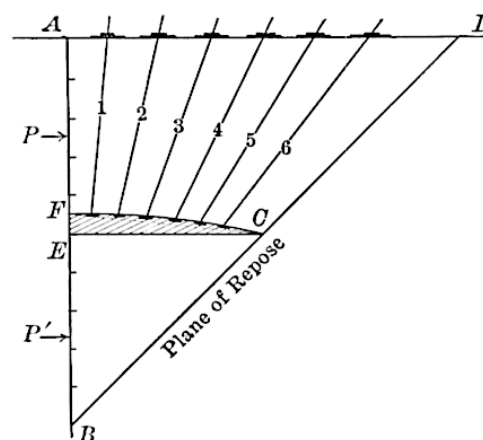


FIG. 22.

It seems that in order to accord our views to the theory of greater pressure at the bottom, we must assume that any given area of the face of a wall is borne on by a prism of material reaching to the top of the bank, and that its pressure is measured by the weight of its cross-section by some multiple of its length. A few moments spent in examining the face of a tunnel drift, or sheeted trench in sharp sand or loam, will convince any one that this cannot be true. In effect, the face of a sheeted trench resists the pressure of what may be termed a series of vertical groined arches, the braces being at the various abutments and the sheeting supporting the more or less loose material between. In no other way can the fact be explained that individual sheeting planks may frequently be removed for a short time

without danger, even when the bracing shows evidence of very heavy pressure; whereas the removal of even a single brace may cause collapse.

A few years ago the writer was called to examine a 44-ft. trench which had collapsed. While it will probably never be known definitely how or where the failure first occurred, it may be of interest to note that within 50 ft. of the break and under conditions apparently similar to those which had existed there previously, the writer found intermediate sheeting planks near the bottom, behind which the pressure was not sufficient to force them out against the rangers, whereas no one could for a moment doubt that there was pressure against the braces or on the sheeting directly behind them. In order, then, that experiments on retaining walls or sheeted faces may be of value, the pressures must be measured against areas absolutely independent of each other; and the writer believes that this can best be done as stated heretofore.

The author [states](#):

“However, if, from any cause, such as insufficient sheeting, the break has taken place over even a part of $A I B E$, the mass, $A F$, above will tend to tip over at the top, giving the greatest pressure on the top braces. This appears to explain the phenomena observed by Mr. Meem and others in connection with some trenches.”

The writer thinks it is unfortunate that the author has not had an opportunity of visiting with him many trenches in which no break had occurred, and yet in which the bracing had to be strengthened continually near the top. This was especially true of a horizontally sheeted and braced shaft of large area. The writer believes that it is not out of place to express the hope that the author may still change his viewpoint, and may eventually regard only as phenomena well-braced trenches which in sand or gravel do not show evidences of heavier pressure near the top.

Referring now to tunnel pressures, the author [states](#):

“If there was no settlement of the earth, $D C F E$, in relation to $D C$, * * * but, as most of the weight of $D C F E$ is carried by the sides, in case of sufficient settlement, the vertical unit pressure, V , on $D C$, will be much less than wh .”

The writer would criticize this view because it brings in the element of settlement as essential to its conclusions, and, therefore, is contrary to his belief that all earth is under stable conditions due to the lateral transmission of thrust due to weight, thereby causing the “cohesive friction” previously noted. This transmission of thrust results in what may be termed “arching stability,” which is unchanged when small areas of hard, dry ground are undermined. When normally dry soft ground is disturbed, however, there is a gradual settlement of tiers of strata, and as those above are, in turn, left without support, they settle on what may be termed a new centering and form themselves into new arches. If these strata are composed of sand and gravel, with little natural cohesion, the action of this settlement will be similar to that observed in an hour-glass and will “work” to the surface in a short time. It should be noted, however, that as the voids below are filled and solidified by the pressure, the lateral thrust of this pressure causes the arching tendencies to be resumed again in each successive stratum, relieving the lower strata of the pressure or shock of fall from those above.

The tables given by the author on [Figs. 17](#) and 18, show that the pressure per square foot on the roof of a 15-ft. tunnel at a depth of 150 ft., for instance, is approximately double that on the same tunnel at a 50-ft. depth. This assumption is not justified by any facts which have ever come under the writer’s notice or have been brought to his attention. The observation of any tunneling operations in soft normally dry ground, or the examination of existing structures, will convince any one that, after a depth of approximately twice the greatest diameter of opening has been reached, it is impossible to tell, by any difference in the pressure, the depth of the

tunnel. While numberless instances could be cited to illustrate this fact, two which have come under the writer's notice may be of interest.

In the case of a 15-ft. tunnel passing the bed of an old underground stream, a considerable amount of ground was lost through the influx of sand which came in with the water. For several days after this the writer examined the surface directly overhead for evidences of settlement, and after some 4 or 5 days he found a hole some 12 ft. in diameter and 8 or 10 ft. deep, which was at once filled in. Had the mass of earth of this area and some 20 to 60 ft. high come down on the timbering suddenly, without any intervening "arch cushion," it would undoubtedly have crushed it; and yet none of the night men had been conscious of even the slightest shock.

In the other case, a heavy rain had caused a large pond to form over the heading of a 15-ft. tunnel, and before it could be drained away it broke through the unprotected face of the heading, virtually filling the whole tunnel with sand for some distance. This, however, had not caused the collapse of any of the bracing, and, before work could be resumed, it was necessary to re-excavate the material. This material was used to fill in the hole caused by the cave-in, and when operations were finally resumed, about 10 days later, the sectional shields, which had remained in position undisturbed, were started with less than the ordinary pressure, as indicated by the gauge on the hydraulic pump.

The writer confidently believes that the assumption behind the reasoning by which the table in [Fig. 17](#) was made is fallacious, and that the fallacy is found in the following quotations:

"[In answering this question](#), it must be remembered that, of the weight of earth directly over the tunnel, all has been transferred to the sides that it was possible to transfer, for the coefficients of friction and cohesion given. We know scarcely anything of the cohesion coefficients, so that the value assumed, $c = 100$ lb. per sq. ft., may not be near the truth. Certainly it must

appear plain from this discussion that the values of c and ϕ must be better known, for all kinds of earth, before reliable results can be attained.”

* * * * *

“In reality, an arch or dome of the earth should be considered in place of the horizontal stratum, but the result is the same, because the same vertical forces act in either case.”

* * * * *

“Referring to Fig. 20, it is evident that the maximum limit of V would be realized if the weight of any horizontal lamina is entirely held up by the friction and cohesion of the sides; * * *”

* * * * *

“As seen, such a state is not exactly realized, but is practically true for great depths.”

Referring to the last quotation, the writer would go further and say that if the assumption is true that the spaces above a tunnel are considered as a series of horizontal layers dependent on the natural coefficient of friction and cohesion (not added to by pressure) to hold them up, that it would appear to be far preferable to calculate always on full pressure to the top than to assume that some of these strata may be sustained by what would appear to the writer to be largely chance conditions.

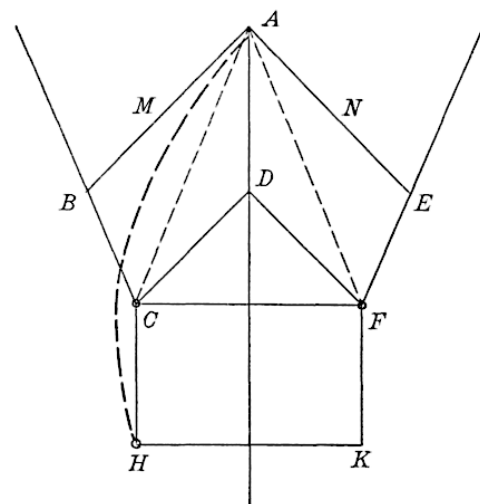


FIG. 23.

It would appear that the author has considered cohesion and friction only as normally found in exposed faces, and as they would be developed between contiguous vertical columns of earth through which pressures were transmitted laterally; and, in tunnels as against vertical faces, he does not appear to have given sufficient weight to the essential factor that cohesion and friction, combined into what the writer has previously termed “cohesive friction,” are increased by the pressure in some definite relation to it.

If, for example, on a tunnel section, $C F K H$, [Fig. 23](#), a centering or core of sand, $C D F$, is assumed, and over this a mass, $A B C D F E$, composed of magnetic particles which cause them to adhere to each other, it is not difficult to conceive that a thickness at the key, $A D$, would be reached where the core, $C D F$, could be removed. The same result, approximately, may be reached by assuming that the mass is sand or earth and is supported by the core, $C D F$. As, however, in this case the lower part of the arch along $C D F$ is composed of loose material, the support of some of it must be provided for in the area where the blending of the arch and the core is indeterminate. This supported area is arbitrarily assumed to be $A C D F A$, or half $M C D F N A$, the point, A , being determined as far as possible by experiment to be at the intersection of the vertical, $D A$, and the line, $C A$, bisecting the angle between that of repose and the vertical. These general deductions, exclusive of the exact determination of the location of the point, A , appear to be borne out by all the experiments previously noted and others in the writer’s papers hereinbefore referred to, and in the author’s observations on grain bins, as noted in the following [quotation](#) from his paper:

“In the many experiments on high grain bins, the enormous influence of the friction of the grain against the vertical walls or sides of the bin has been observed. In fact, the greater part of the weight of grain, even when running out, is sustained by the walls through this side friction. This furnishes another argument for including wall friction in a retaining-wall design.”

Not only is this “an argument for including wall friction,” but it seems to prove that this friction is increased relatively to the pressure, and that under stable conditions coherence is also induced by the pressure and friction.

The writer is much gratified to find that the author concurs in the view that the area of water pressure is reduced in subaqueous tunnels and other submerged structures in sand or earth, and he concurs heartily with the author that experiments on a large scale, to determine the values of this reduction definitely with relation to the various materials, will be of the greatest value to the Profession.

WILLIAM CAIN, M. AM. SOC. C. E. (by letter).—The writer is gratified by Mr. Worcester’s words of commendation. The walls or boards subjected to earth pressure were of various inclinations, and the surface slope of the earth was equally varied. A theory which stands the test of experiments in such variety seems to be pretty well established. If the various theories that have been proposed from time to time were subjected to this test, how many would survive? And yet no theory can claim to be a practical one unless it is found to agree fairly well with experiments. Mr. Worcester seems to think that the effect of time on retaining walls ought to be included. The effort was made to do this, by using a factor of safety and by multiplying only the normal component of the thrust on the wall by this factor, taken as 3 for ordinary cases; where the effect of frost is decided, the factor should be increased, and the back of the wall, for say 3 or 4 ft. down from the top, should be sloped forward to allow the earth, in the expansion incident to freezing, to push its way up the inclined plane corresponding.

If railway trains pass near a retaining wall, their weight should be replaced by an equal weight of earth, which is regarded as dead weight in computing the thrust. Vibration probably increases the thrust, and this increase moves the top of the wall over slightly, on account of the yielding of the earth foundation about the outer toe. On account of the imperfect

elasticity of earth, this deformation may remain and increase in time, and thus lead to the ultimate failure of the wall. This lack of spring, or recovery, in the earth foundation, is probably the main cause of the increased leaning of walls with time. The remedy is to build a foundation course of masonry, projecting in front of the wall, of such width that the true resultant on its base shall pass through its center. The base, too, should be inclined, in order to prevent sliding. Of course, efficient drainage must be secured by the use of weep-holes and perhaps drains back of the wall.

Mr. Worcester is of the opinion that the friction against the back of the wall is not a permanent feature, and suggests the Rankine formula as possibly a better one for design. For a surcharge of sufficient inclination, and especially when it slopes at the angle of repose, the Rankine thrust involves more friction at the back of the wall than the method illustrated in [Fig. 15](#), where only one-third of the friction is used for ordinary cases; but, even granting, for the sake of argument, that at some time this friction is null and that subsequently rains and vibration cause an increased thrust, then the top of the wall moves over slightly, the earth will again get its frictional grip on the wall, so that this friction is always exerted when required for stability.

It is, perhaps, customary to design a wall so that the resultant on its base shall pass one-third of the width of base from the outer toe. This procedure gives very different factors of safety, as hitherto defined, for different types of walls. The writer's method aims to give equal security to all ordinary walls, by using a constant factor of safety.

The writer is gratified that Mr. Meem has again recorded some of his valuable experiences, but regrets that he cannot regard some of his theories as convincing. With regard to the center of pressure on a retaining wall backed by fresh earth, Mr. Meem maintains that the intensity of pressure increases from the bottom upward, so that the center of pressure lies above

the horizontal plane drawn at mid-height. This view has been shown to be untrue by experiment. Thus:

(1) Leygue, in the experiments referred to on [page 420](#), found the value of the moment of the earth thrust about the inner toe, and also determined the plane of rupture. Using the corresponding wedge of rupture, the writer computed the normal component of the thrust. On dividing the moment by this, the distance of the center of pressure from the base was found to be, as an average, for all experiments on sand, 0.34 of the height, and for millet seed, 0.405 of the height.²⁵

(2) The easily made experiment of Mr. Gifford²⁶ on the deflection of a cardboard retaining wall shows that the resultant thrust lies nearer the base than the top of the wall.

(3) All the experiments discussed on [pages 407–427](#) agree with the latter statement.

(4) The resultant earth thrust on a wall must approach indefinitely water thrust as ϕ approaches zero. The latter is known to act at one-third of the height above the base.

(5) If a triangular wedge of rupture is assumed, it follows inevitably that the unit pressure increases with the depth, and that the resultant acts at one-third of the height above the base. This follows because the total pressure then varies with the square of the height, as in water pressure.

(6) Let the contrary be assumed—that the pressure increases from the base upward—then, in a great depth of earth, the pressure at the top would be enormous enough to crush the hand if thrust in the earth. As everyone knows, the pressure on the hand is very slight, and this shows the absurdity of the hypothesis.

It may be stated now that the proposed experiment, referring to [Fig. 22](#), would not prove Mr. Meem's contention. For, if the earth below FC was

removed, the thrust on AF would have to be sufficient to prevent the whole mass, $AFCD$, from descending, which is far greater than P , which balances only the thrust of the wedge of rupture, the inclined base of which passes through F .

A conclusive experiment could be made on a high retaining wall, backed by sand or grain (not in a bin, but unconfined except by the wall) after Jamieson's manner in the case of grain bins, by inserting the rubber diaphragms, etc., at various points from the top down, and measuring the pressures.

In respect to the distribution of pressure, the theory of the sheeted trench differs materially from that of the retaining wall. Much confusion has arisen from confounding them. On that account, and to meet many interesting points made by Mr. Meem, the writer will give a thorough discussion of retaining walls and sheeted or unsheeted trenches, backed by coherent earth.

For an unlimited mass of level-topped earth, having both friction and cohesion, but ignoring the cohesion along the vertical plane, DE , Fig. 24, it was shown in the Appendix that the horizontal pressure on a vertical plane, AC , is that due to a certain prism, $ADEC$, and that its total amount, in pounds, is,

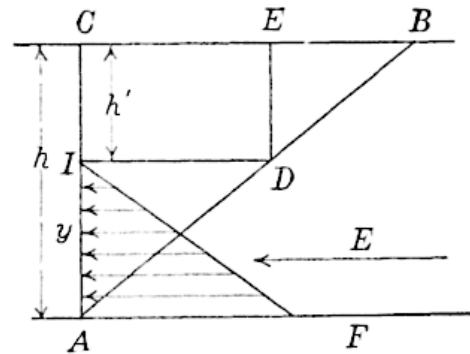


FIG. 24.

$$E = \frac{1}{2} w y^2 \tan^2 \left(45^\circ - \frac{\phi}{2} \right) \quad (7)$$

where w is the weight in pounds of 1 cu. ft. of the earth, c is the cohesion of the earth, in pounds per square foot, $y = h - h' = AC - IC = AI$,

$$h' = IC = \frac{2c}{w} \frac{\cos \phi}{1 - \sin \phi} = \frac{2c}{w} \tan \left(45^\circ + \frac{\phi}{2} \right). \quad (8)$$

The plane, AD , is found to bisect the angle between the natural slope and the vertical when E is horizontal, as shown by [Fig. 24](#).

The value of E is given just below [Equation \(4\)](#) in the Appendix. It acts at $\frac{1}{3} y$ above the base, since the distribution of stress on AI is linear.

It follows, because [Equation \(7\)](#) is the usual one for the thrust of non-coherent earth of depth y , that the total horizontal stress on AC of the earth endowed with cohesion, is exactly the same as that due to the same earth, but devoid of cohesion, having a free horizontal surface, ID , extended indefinitely in both directions, at a depth h' , as given by [Equation \(8\)](#), below the original free surface. The theory ignores any possible cohesion acting upward along DE , or any tension in the mass that may possibly drag down part of the wedge, EDB , and thus decrease the thrust, E .

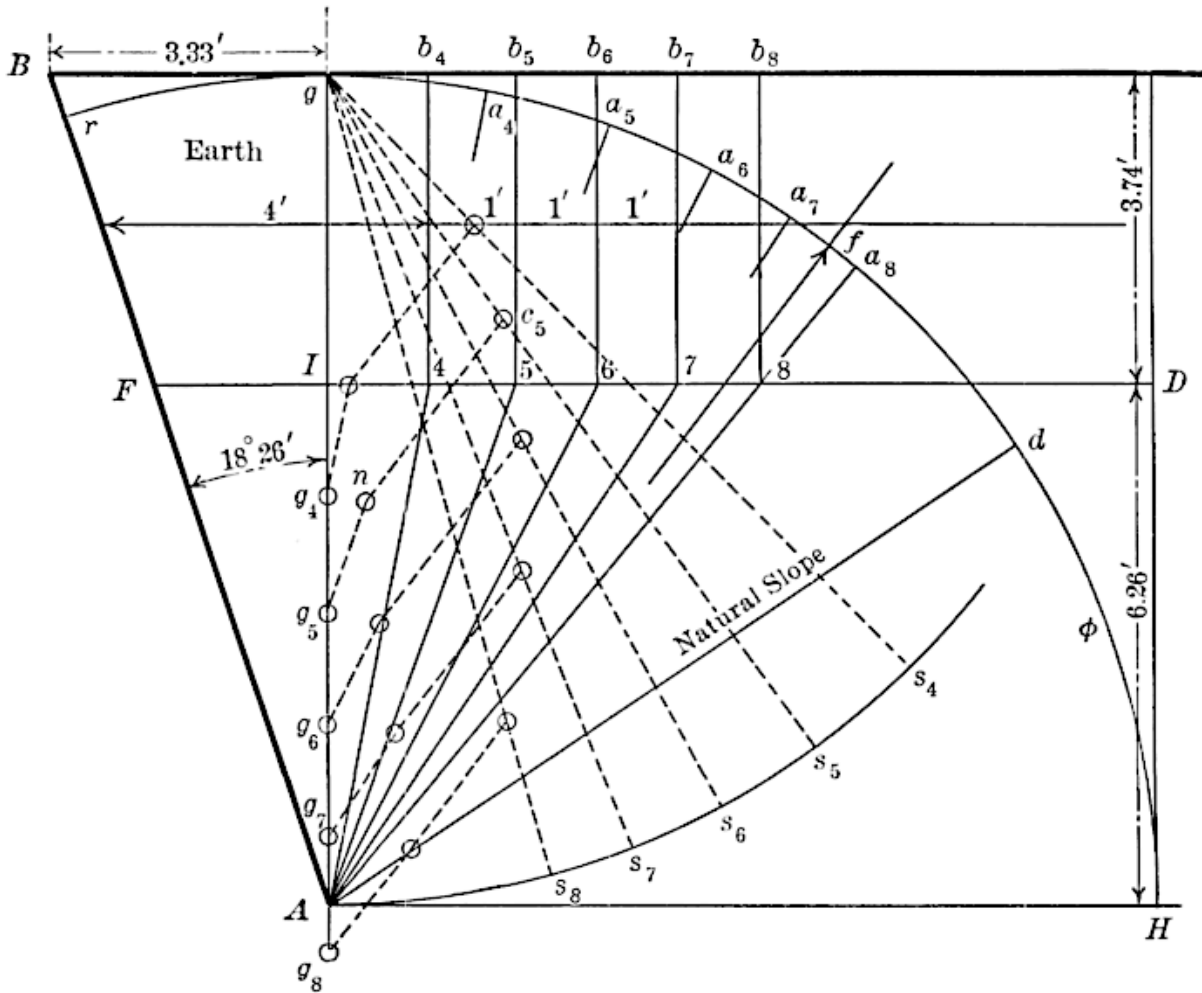
The formula is thus seen to give a thrust greater than the true one, or what may be called an upper limit. To realize the hypothesis more clearly, it may be said that if a vertical crack in the earth is assumed along DE , the resulting value of E will be the same as that given by [Equation \(7\)](#). It is a fact of observation that sometimes earth which has been saturated and then dried out, cracks along one or more vertical planes. This indicates tension in the mass, which is overcome, however, at certain points (only) and thus vertical cracks appear.

In the construction of [Fig. 11](#), the full friction and cohesion which can be exerted on the length, AB (of [Fig. 24](#)), is supposed to be exerted. This construction then gives a lower limit to the thrust. As to which hypothesis will lead to the most probable value, it may be observed that the broken line of rupture, ADE , [Fig. 24](#), is nearer the true curved line of rupture (which is assigned both by theory and the facts of observation) than the straight line,

AB ; hence, the hypothesis leading to [Equation \(7\)](#) seems to be the more probable one.

In the case of the open trench, suppose AC to be a vertical side of the trench; if A lies below the level of ID , and a crack exists along DE , then, undoubtedly, the mass, $ADEC$, will move to the left, because there is an unbalanced force, E , to cause the motion.

The vertical height of an unsupported bank, where vertical cracks occur of depth h' , will then be that given by [Equation \(8\)](#), which is one-half the value usually given. This is generally an extreme lower value; for any earth endowed with much cohesion must be capable of exerting tension. If the tension exerted is sufficient to drag down DEB (which can stand unsupported), that is, if the wedge, ABC , acts as a whole, then the free unsupported height will be double that given by [Equation \(8\)](#). This is evidently an extreme upper limit, perhaps rarely attained; for vertical cracks have often been observed to precede a fall of earth into a vertical trench. For $w = 100$, $\phi = 35^\circ$, and $h' = 23$ ft., [Equation \(8\)](#) gives $c = 600$, whereas the usual equation gives $c = 300$ lb. per sq. ft. The true value is possibly between these two extremes.



Recurring again to earth pressures, the discussion pertaining to Fig. 24 suggests the following modification of the graphical method of Fig. 11 to adapt it to finding the pressures of coherent earth against retaining walls, for the case supposed above. In Fig. 25, let AB be the inner face of a retaining board or wall, which is backed by earth with a horizontal surface Bb_8 . The vertical height of the wall is 10 ft. and the physical constants are assumed as follows: $\phi = 33^\circ 41' = \phi'$, $w = 100$, and $c = 100$. From Equation (8),

$$h' = \frac{2c}{w} \tan \left(45^\circ + \frac{\phi}{2} \right) = 3.74 \text{ ft.}$$

Now lay off, vertically downward from the free surface, a distance, $h' = g I = 3.74$ ft. (to scale) and through I draw a line, FID , parallel to the top surface. By Equation (7), the thrust on any vertical plane as b_4 4, b_5 5, ..., is null, and any tension and cohesion that may actually exist along any of these planes will be ignored. This is equivalent to supposing vertical cracks along these planes, which leads to an increase of the thrust and is thus on the side of safety.

The weights of the successive trial prisms of rupture, $A4 b_4 B$, $A5 b_5 B$, ..., are now laid off to scale, along some vertical, as $g A$. The successive weights are represented by the vertical lines, $g g_4$, $g g_5$, ..., $g g_8$. The points b_4 , b_5 , ..., were taken 1 ft. apart; hence the area, $A4 b_4 b_5 5 =$ the area $A5 b_5 b_6 6$, etc.; so that, after computing the area $A4 b_4 B$, the areas $A5 b_5 B$, etc., can be found by successive additions. These areas, multiplied by 100, give the weights, in pounds, of the successive prisms of rupture for 1 ft. length of wall. As before, two arcs are drawn with the same radius ($A g$), with centers A and g and having laid off the angle, $HAd = \phi$, the chords $A s_4$, $A s_5$, ..., are laid off equal to chords $d a_4$, $d a_5$, The lines, $g s_4$, $g s_5$, ..., now make angles with the normals to the planes, $A4$, $A5$, ..., respectively, each equal to ϕ . Also, lay off the chord, $d f = g r$; then $A f$ gives the direction of the reaction of the wall, AB (directly opposed to the earth thrust), inclined at the angle, $\phi' = \phi$, to its normal.

Next, measure, to the scale of distance, the length, in feet, of any line of rupture, as $A5$. Multiply this number by $100 = c$, to get the force of cohesion, in pounds, acting up along $A5$, and lay it off, to the scale of force, from g_5 , on a line parallel to $A5$, to n . Similarly, lay off the cohesive forces, acting along $A4$, $A6$, ..., at g_4 , g_6 , ..., and from points, such as n , lines are drawn parallel to the direction of the thrust, $A f$, to the intersections with the corresponding rays, $g s_4$, $g s_5$, Suppose $n c_5$ to prove the greatest of these segments, then $n c_5$, to the scale of force, gives the earth thrust on AB , in pounds. In the present instance, the plane of rupture lies midway

between A_5 and A_6 , and the corresponding thrust is 1 440 lb. For purposes of illustration, regard A_5 as the plane of rupture; then the forces acting on the prism, $A_5 b_5 B$, are represented by the sides of the closed polygon,

$$g g_5 n c_5 g;$$

$g g_5$ representing its weight, $g_5 n$, the cohesion acting up along A_5 , $n c_5$ the reaction of the wall, and $c_5 g$ the resultant of the normal reaction, N , of the plane, A_5 , and the friction, $N \tan \phi$, acting along it. As stated, the actual plane of rupture is found (by drawing one or more additional trial planes) to lie midway between A_5 and A_6 .

If it should be deemed desirable to include the cohesion acting upward along the vertical planes, $b_4 4$, etc., it is very readily done. At each point, such as n , draw vertically upward a line of length (to scale of force) $= c h' = 100 \times 3.74 = 374$ lb., to represent the force of cohesion acting along the corresponding plane. From the extremities of such lines, parallels to $A f$ are drawn to the intersections with the corresponding rays, of the type, $g s$. As before, the greatest of these lines represents the thrust on AB . Its amount, in this instance, is 1 240 lb. It is seen, especially when $h' = g I$ is one-half of $g A$ or more, that the change in thrust is quite appreciable.

Lastly, if each plane, as A_5 , is extended to the surface and considered to offer full cohesive and frictional resistances throughout its whole extent, the construction of [Fig. 11](#) applies, and gives a thrust of 1 220 lb., practically the previous amount.

As mentioned before, in connection with [Fig. 24](#), this supposes sufficient tension in the mass to drag down the triangle, DEB . Suppose, now, $A g$ to be the face of a trench which will just stand unsupported for the height, $A g$, when the cohesion along $b_4 4$, $b_5 5$, ..., is included. Then, if, from the drying out of the earth, causing contraction near the surface and possibly changes in c and ϕ , cracks occur along some of these planes, the resistance to motion of the corresponding prism of rupture is decreased, and

the mass will move down, unless, in the meantime, sufficient bracing has been put in. This well illustrates what constructors tell us of the importance of getting in well-keyed-up bracing in time to prevent any crack from developing, which, as we have seen, largely increases the thrust. Even when cracks do not appear, a heavy rain, shortly after a trench is opened, is a frequent cause of falls; evidently because c and possibly ϕ have been very materially decreased.

To compare the results of this very general graphical method with those given by analysis, as expressed by [Equations \(7\)](#) and [\(8\)](#), let AB be taken vertical or coinciding with $A g$, regard the thrust on $A g$ as horizontal, and neglect any cohesion on the vertical planes, $b_4 4$, etc. Then it is found, for the given constants, that the graphical method gives exactly the thrust (560 lb.) obtained from [Equation \(7\)](#). This result was foreseen.

It is likewise interesting to know, if the thrust on $A g$ is taken as parallel to $A d$, or making the angle, ϕ , with the normal to $A g$, that the thrust on $A g$, as given by the construction, is again found to be the same (510 lb.) as that given by a well-known formula for the thrust on the plane, AI , for earth devoid of cohesion and having a free surface, ID . The importance of these conclusions lies in this: that for the wall vertical, the earth surface horizontal, the earth thrust being horizontal or otherwise, the shorter method is available. If preferred, the thrust on the wall, AI , for the earth devoid of cohesion, with the free surface, ID , for either direction of the thrust, can be evaluated by the graphical methods ([page 404](#)) hitherto given.

Since this paper was written, Résal's "*Poussée des Terres*," *Deuxième Partie*, on Coherent Earths, has appeared.²⁷ In it the author gives an exhaustive discussion of lines of rupture for a great number of cases. The equivalent of [Equation \(7\)](#) is found for the case of the horizontal pressure on a vertical plane when the free surface of earth is horizontal; but it was found to be impracticable to derive a formula for the earth thrust for the general case of the earth surface sloping at an angle to the horizontal, the

wall being either inclined or vertical. In fact, for such cases, the intensity of the earth thrust at any depth is not a linear function of the depth, as obtains in the case shown by [Fig. 24](#). Hence Résal resorts to the following approximation: Conceive a line drawn parallel to the surface, at a depth, h' (as given by [Equation \(8\)](#)), below it, and regard this line as the free surface of non-coherent earth of the same specific weight and angle of repose as the given earth; compute the thrust against the wall for such earth, devoid of cohesion, by methods pertaining to such earth; the thrust thus found is assumed to be approximately the true thrust on the wall for the original coherent earth. It is proper to state that Résal rejects the sliding-wedge theory for non-coherent earth, and uses a method of his own, which involves elaborate tables given in his book. The wedge theory is admittedly imperfect, mainly because the surface of rupture is a curve, but we have seen that it agrees with experiments on model walls or retaining boards, when properly interpreted, and it will be used, as before, in computing the earth thrusts, E' , below, for earths devoid of cohesion. The graphical method has already been indicated.[28](#)

In [Table 6](#) comparative results are given for various cases, including those already examined. Each retaining board was supposed to be 10 ft. high, the earth to have a natural slope of 1 on $1\frac{1}{2}$, and to weigh 100 lb. per cu. ft.

α = the inclination of the inner face of the wall to the vertical;
 $\tan \alpha$ will be considered as positive when the top of the wall leans toward the earth, otherwise negative;

i = the angle the surface of the surcharge makes with the horizontal;

c = the cohesion, in pounds per square foot;

ϕ' = the assumed inclination of the thrust to the normal to the face of the wall;

E = the thrust for coherent earth, found by the writer's construction, the general case being shown in Fig. 25; in this the cohesion along the vertical planes was omitted;

E' = the thrust for non-coherent earth, by the usual methods, where the new surface of the non-coherent earth is supposed to be a plane parallel to the original surface, and at a distance h' , as given by Equation (8), below it; at this depth, it can be rigorously proved, for any surface slope, that the thrust is zero.

TABLE 6.

Case.	$\tan \alpha.$	$\tan i.$	c , in pounds per square foot.	ϕ'	E , in pounds.	E' , in pounds.
1.	$-\frac{1}{3}$	0	100	ϕ	1 440	880
2.	0	0	100	0	560	560
3.	0	0	100	ϕ	510	510
4.	0	$\frac{1}{2}$	100	ϕ	660	750
5.	0	$\frac{2}{3}$	100	ϕ	880	1 630
6.	$+\frac{1}{3}$	0	50	$18^{\circ}26'$	240	490

It is seen, by comparing the values of E and E' , in Table 6, that, except for Cases 2 and 3, where the coincidence in the results has been already noted, the thrusts by the two methods differ very widely, hence the second

method must be rejected, as in some cases undervaluing the thrust, and in other cases overvaluing it. In Case 5, where the surcharge slopes at the angle of repose, the large excess is due principally to the ordinary theory for computing E' , involving an infinite plane of rupture, as hitherto noted.

In Case 6, where the top of the wall leaned toward the earth, c was first assumed equal to 100, but it was found to give no thrust against the wall; which means that the earth would stand unsupported at the slope $\frac{1}{3}$ or with the face making an angle of $18^{\circ}26'$ with the vertical, when this face was 10 ft. high. Hence, a second trial was made, with $c = 50$, with the results shown. It will be noticed that ϕ' was assumed as $18^{\circ}26'$, so that the thrust was taken horizontal. In fact, Résal asserts that the thrust against such a leaning wall, makes a less angle than ϕ with the normal to the wall. According to his tables, ϕ , for this wall cannot exceed $20^{\circ}40'$. As the wall approaches the vertical, ϕ' , approaches ϕ , the exact value given for a vertical wall. By comparing the thrusts, E , for Cases 1 and 6, the economy of using the latter type of wall is so apparent that it is astonishing that constructors do not adopt it oftener.

The general conclusion to be drawn from Table 6 is that, except for Cases 2 and 3, the general graphical method of Fig. 25 must be used for accuracy. If applicable to the case in hand, the cohesion along the vertical planes, b_4 4, etc., can be included with very little additional labor. The graphical treatment given is so general and the theory involved is so apparent to the eye, that it seems to commend itself as a practical treatment of a

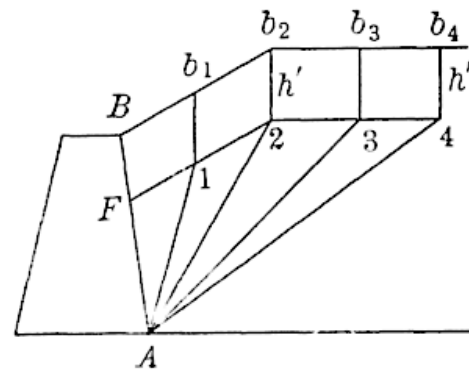


FIG. 26.

The general method illustrated in Fig. 25 can also be applied to the surcharged wall of Fig. 26. Here, the lines, $F2$ and 24 , are drawn parallel, respectively, to Bb_2 and b_2b_4 , and vertically, $h' = 2b_2 = 4b_4$ ft. below them. The trial prisms of rupture are of the type, $A3b_3b_2B$. Their weights are laid off as before, on the line, gg_8 , of Fig. 25, and the further construction is exactly as there indicated. It is assumed here that the slope of Bb_2 does not exceed the natural slope, for, if it does, the construction is somewhat altered.

It may be asked whether the construction of Fig. 25, where the cohesion on the vertical planes, $4b_4$, etc., is omitted, if applied to the experiments on rotating boards, will appreciably alter the results given in Table 3. The answer is no; for, when $c = 1$, $w = 80$, therefore $h' = 0.046$ ft., or is very small. In fact, as in the experiments, no cracks were formed, the cohesion on the vertical planes should be included, which would lead to the results given.

After the thrust, for earth endowed with cohesion, has been computed, the next question is, at what point on the retaining wall does it act? This can be answered at once, when the inner face of the wall is vertical and the earth surface horizontal, for then the earth thrust acts at $\frac{1}{3}$ of the height, AI , Figs. 24 and 25, as hitherto proved. The case is not so simple when the wall is inclined, either toward or from the earth. In Fig. 25, the thrust, $n c_5$, on the wall, AB , must be decomposed into horizontal and vertical components. The horizontal component is the resultant of the horizontal forces acting from F to A which may be assumed to follow the linear law; hence this component will act on AF at a point, distant $\frac{1}{3} AF$ from A , going from A to F . The vertical component, similarly, will be regarded as acting on AB , at a distance $\frac{1}{3} AB$ from A . The same approximate rule is suggested when AB lies to the right of the vertical, $A g$.

When the earth surface slopes at the angle of repose, the distribution of stress is not linear, but more like that shown in [Fig. 10](#). As the equation of the corresponding pressure curve cannot be found, an approximation only can be suggested. When $\frac{h'}{h}$ is (say) less than one or two tenths, the factor, $\frac{1}{3}$, above, can be used; but as $\frac{h'}{h}$ increases toward its limit, 1, the factor increases toward some unknown limit. Probably it does not exceed 0.4 even for [Fig. 26](#), and, for want of accurate knowledge on the subject, it may be taken at 0.4 as a rude approximation.

Finally, when the surface slope is less than the natural slope of the earth, then as it decreases from $\tan \phi$ to zero, the factor should. decrease from the extreme value (say 0.4) to $\frac{1}{3}$. Résal takes this factor at $\frac{1}{3}$ for all cases, which is certainly not on the safe side.

The case of the retaining wall which receives the active thrust of the earth has been hitherto examined, and next the case of the braced trench will be discussed. As the trench (having vertical sides) is dug, the usual sheeting, rangers, and bracing are put in and the bracing is kept well keyed-up, so as to exert an active pressure on the earth. To illustrate the theory, in [Fig. 25](#), let $A g$ represent the vertical side of the trench, the earth extending only to the right of $A g$. Then from points such as n , draw horizontal lines to the intersection with the corresponding gs ; the longest of these lines, to the scale of force, will represent the total force that must be exerted by all the braces, per foot of length of trench, to prevent any motion of the mass. As has been seen, this force is given by [Equation \(7\)](#). If a still greater force is exerted by the braces, less than a certain value which would just cause motion of the earth up some plane of rupture, stability is completely assured.

As a numerical illustration, assume a trench 40 ft. deep and 15 ft. wide, the constants for the earth being, $\phi = 33^\circ 41'$, $c = 100$, $w = 100$. Then, by

Equation (8), $h' = 3.74$ ft. and $y = h - h' = 40 - 3.74 = 36.26$ ft. Hence, by Equation (7), the least force the braces must exert per linear foot of trench is,

$$\frac{1}{2} w y^2 \tan^2 \left(45^\circ - \frac{\phi}{2} \right) = 50 (36.26)^2 \times 0.2866 = 18844 \text{ lb.}$$

Suppose the braces to be 10 ft. apart horizontally, and that there are six braces in the same vertical plane. The least force that the horizontal braces must exert on 10 lin. ft. of trench is 188 440 lb., and if each carries the same stress, the force to be exerted by one, is $\frac{188440}{6} = 31407$ lb. Assuming a unit stress of 800 lb. per sq. in. for an 8 by 8-in. wooden brace, 15 ft. long, it is seen that one brace can safely exert a force of 51200 lb., which is greater than the least amount required, as should be the case to allow for changes in c and ϕ due to heavy rains. To meet such contingencies, 10 by 10-in. braces are suggested. Of course, very little is now known as to the coefficient, c , but, from the observed heights of trenches which have stood without sheathing, it is probable that values of c from 100 to 300 lb. per sq. ft. can be counted on for most trenching. For the present, the only safe way is to be guided by experience, such as has been elicited by Mr. Meem's paper, "The Bracing of Trenches and Tunnels, With Practical Formulas for Earth Pressures."

As to the exact distribution of the stresses, theory cannot speak definitely, for the conditions are different from those in the case of the retaining wall, the passive resistance of which opposes the active earth thrust. For the braced trench, the earth, at first, simply resists the active pressure exerted by the braces, when first put in and keyed-up tight, particularly on that upper portion where the active earth thrust is nothing or very small. As the construction proceeds, the braces will receive more and more of the active earth thrust, which necessarily increases with the depth of trench. In fact, the distribution of stress, indicated by the arrows in Fig. 24, although true in the case of a retaining wall, where the earth has

been deposited behind it, is not necessarily or generally true in a sheeted trench, because of the manner of its construction. Thus, in digging a trench, the bracing is put in at intervals, but when a brace is inserted near the bottom of the trench, the digging is continued for several feet without bracing, until a depth is attained at which it is thought best to insert another brace. Before the latter is put in, the unprotected face of the trench, say 6 ft. in depth, can exert no pressure, as it would in the case of the retaining wall. The thrust that would be exerted, for this area, on the supposed wall, does not exist for the unsupported face of the trench, for the full horizontal thrust of the earth for the whole depth has been taken up by the braces above the unsupported area. This state of affairs is characteristic of the work as it proceeds, the lowest brace that is put in at first carries only the stress due to the keying-up, but takes more and more stress as the excavation proceeds. It can thus very well happen that the upper or the middle braces may receive more stress in the end, than the lower braces. In fact, this was asserted to be generally true, for well-drained material, by many engineers in the valuable discussion on Mr. Meem's paper referred to. Other engineers advised caution in accepting this view, and asserted that in wet or saturated ground the lower braces were most severely stressed. If it were possible to force a board of a size equal to the length and depth of a trench, vertically downward, excavate and brace, all in a millionth of a second, then one can conceive that the distribution of stress shown in [Fig. 24](#) might be realized; but, for trenches as actually constructed, the distribution of stress in the earth mass is very much altered from this, though it would appear that the total earth pressure would be given, at least approximately, by the construction of [Fig. 25](#), or by the use of [Equation \(7\)](#).

The writer hopes that Mr. Meem may agree to the foregoing explanation, for the subject of trenches was entered into in great detail, in order to explain, if possible, all the facts, as presented by many engineers who held very diverse views about the explanation of them. Mr. Meem's [Fig. 22](#) can illustrate the writer's view: if bracing only extends, say, from *A*

to E , then the braces must exert sufficient horizontal thrust to prevent the descent of the wedge, say $AB3$, if this gives the maximum thrust for the coherent earth.

Turning now to pressures on tunnels, the writer is pleased to note that Mr. Meem has recorded some more of his experiences concerning them. From lack of proper knowledge of the so-called constants, k , c and ϕ , such experience is an aid in leading to more probable values.

As the writer proceeds in this investigation, he hopes to make clear the conception of arch action alluded to, and will derive the limiting values of V and L in the simple manner outlined briefly in the Appendix. Before considering these matters, however, it may be well to call attention to the fact that a vertical prism of earth can be held up by the cohesion of its sides alone when c is large enough. Take the vertical prism of depth h and cross-section area A , [Fig. 20](#), its weight being $w A h$. In a long tunnel, two sides only can be counted on to furnish cohesive forces. Call U the perimeter available in any case. Then the cohesive force is $c U h$, and this alone (without reference to any friction that may act) will support the prism when, $c = w \frac{A}{U} = w R$. In the long or completed tunnel, $R = \frac{b l}{2 l} = \frac{b}{2}$ (see notation of Appendix); but if we take a short section at the heading, say

$l = \frac{b}{2}$, then support can be derived from three sides, and $R = \frac{\frac{b^2}{2}}{2 b} = \frac{b}{4}$. Thus if we assume the horizontal cross-section of the prism to be 15 by 7.5 ft., $c = 90 \times \frac{15}{4} = 338$ lb. per sq. ft., where $w = 90$ lb. per cu. ft.

Thus, such a vertical prism of any height, at the heading, can be sustained by a cohesion of 338 lb. per sq. ft. acting on three of its sides. This refers to a tunnel 15 ft. wide. For the completed tunnel 15 ft. wide, the part vertically over the tunnel can be supported if the two sides can furnish double this unit cohesion. The term $(w R - c)$ will appear in an equation mentioned later. It is seen now that when this term is zero, cohesion alone can sustain

the prism, and the pressure reduces to zero. When this term is negative, it indicates that too large a value of c has been assumed, for stability is assured for $c = w R$.

In reference to Fig. 16, it was stated that a series of superposed arches or domes were assumed, but that since the reactions of the horizontal laminae of Fig. 20 and corresponding arches were the same, the former were substituted for convenience. If each arch has the vertical thickness, dy , it will have the same volume and weight, $A W dy$, as the horizontal lamina. The reactions marked on Fig. 20 can be assumed to be those of the corresponding arch, and the resultant of the horizontal and vertical reactions represents the thrust of the arch at the sides; its direction will be that of the tangent line of the arch. When c is small, the horizontal reaction is much greater than the vertical one, and the arches are all very flat. Now, considering the whole series of arches, it is plain that the greatest or limiting value of V (call it simply V) would be realized if the weight of any arch is entirely held up by the friction on the sides (induced by the lateral thrust) and the cohesion acting there; for then this same condition of affairs would exist for all lower arches,²⁹ and thus V is transmitted vertically downward, unchanged, to the tunnel. Stating this condition in algebraic form,

$$A w dy = (L U \mu + U c) dy.$$

This equation strictly holds for large values of y , but it is practically true for much smaller values. For such values, h' is quite small compared with y and can be assumed as equal to zero, whence V' will also be taken equal to zero and the usual bin formula, $L = V k$, written.³⁰

On substituting this value in the equation above, reducing and placing $R = \frac{A}{U}$, we obtain,

$$V = \frac{1}{k\mu} (w R - c) : L = k V = \frac{w R - c}{\mu} \quad (9)$$

where, $\mu = \tan \phi$.

In the Appendix, the writer assumed, $k = \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$, basing this value on the results pertaining to a smooth steel bin, 1 ft. in diameter, filled with sand. From the experiments of Jamieson on large wooden bins, about 12 by 12 ft., filled with wheat, k was found to be 0.60. Janssen determined k experimentally to be 0.67. For wheat, Jamieson gave $\phi = 28^\circ$; whence $\tan^2 \left(45^\circ - \frac{\phi}{2} \right) = 0.361$. His experimental value was $\frac{10}{6}$ of this; hence, for lack of definite data, there will be assumed for earth,

$$k = \frac{10}{6} \tan^2 \left(45^\circ - \frac{\phi}{2} \right). \quad (10)$$

From the last three equations, the values of V and L have been computed for various values of c , b , ϕ , and R , and are given in [Table 7](#).

TABLE 7.

c, in pounds per square foot.	b, in feet.	ϕ	V, IN POUNDS PER SQUARE FOOT.		L, IN POUNDS PER SQUARE FOOT.	
			$R = \frac{b}{4}$	$R = \frac{b}{2}$	$R = \frac{b}{4}$	$R = \frac{b}{2}$
(1)	(2)	(3)	(4)	(5)	(6)	(7)
100	15	30°	740	1 790	410	1 000
100	30	30°	1 790	3 900	1 000	2 160
100	15	45°	830	2 010	240	570
100	30	45°	2 010	4 370	570	1 250
400	15	30°	0	860	0	480
400	30	30°	860	2 960	480	1 640
400	15	45°	0	960	0	270
400	30	45°	960	3 320	270	950

The pressures given in Columns 5 and 7 are intended to apply to a long section of a tunnel, those in Columns 4 and 6 refer to a short section about

the heading. The values for $c = 100$ are intended to apply to what Mr. Meem styles “soft normally dry ground,” and it is hoped that he may approve the figures, as they are somewhere near his own. The coefficient, $c = 400$, refers to hard consolidated ground. Here the pressure is 0 at the working faces of the 15-ft. tunnel. [Fig. 27](#) shows the variation in V for $c = 100$, $b = 15$, $\phi = 30^\circ$, for some of the larger values of y , as obtained by the revised formula given in the foot-note. It will be observed that the above demonstration for finding the limiting value of V , is perfectly independent of Janssen’s formula. In it the relation, $L = V k$, is only assumed to be true for this one value of V , and k need be determined by experiment only for this value. The result is thus general, no matter how k varies for other values of V . A glance at all the diagrams,[31](#) giving the experimental values of V and L for various depths, will show that k is far from being a constant for varying depths, though the assumption is found to lead to practical results, as obtained from Janssen’s formula. The experiments of Jamieson on 12 by 13½ by 67½-ft. wheat bins, and of Bovey on 12 by 14 by 44 ft. 10-in. bins, both of wood, indicate that the maximum pressure, V , is realized, practically, for heights of about four diameters. Pleissner’s experiments on a wooden bin, 11.51 by 8.20 ft., show four and a half diameters, and Luft, for a concrete bin, 23 ft. in diameter, gives, say, three diameters, for the height corresponding to maximum V .

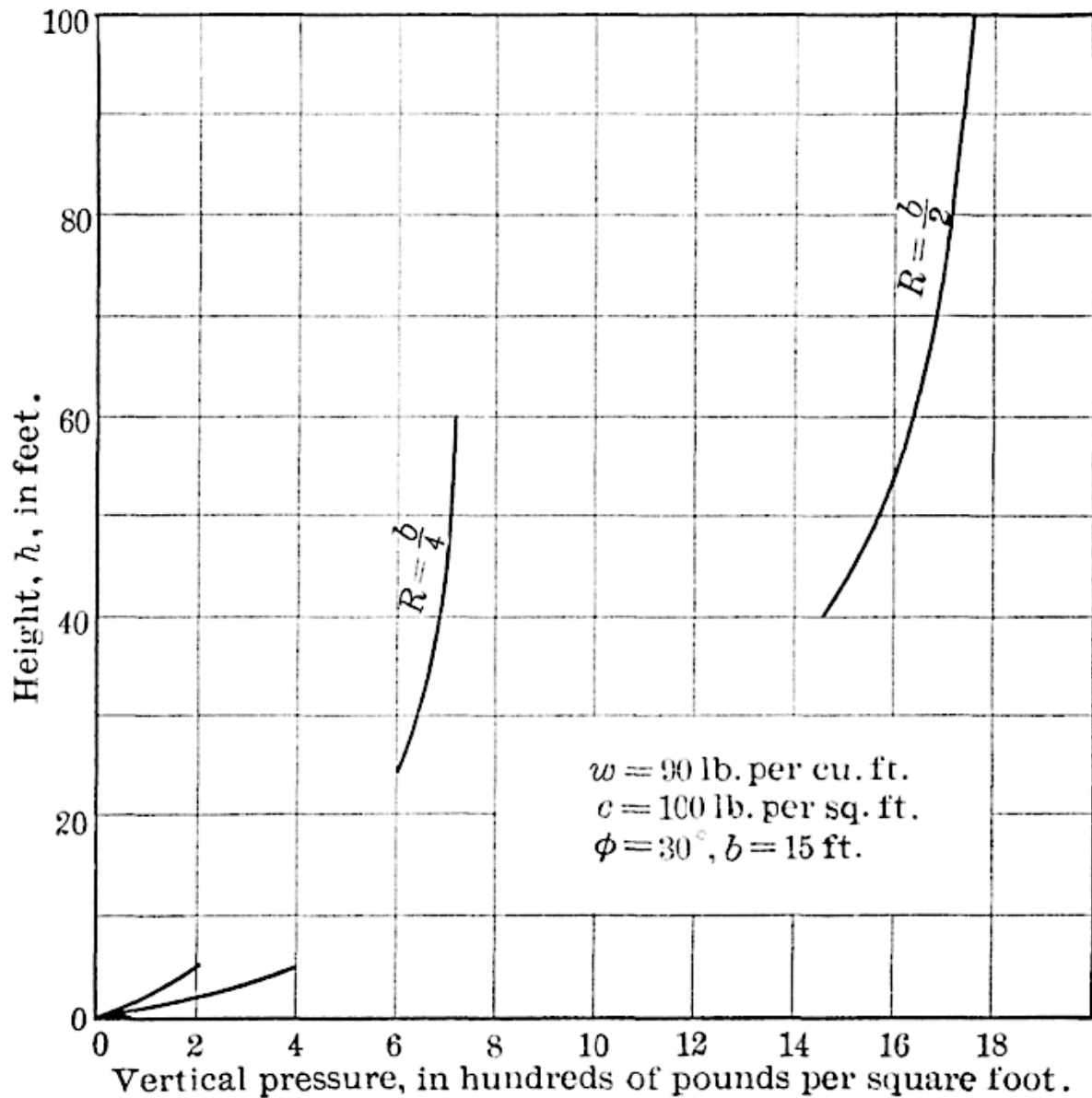


FIG. 27.

These are wide variations, resulting from variations in k and the coefficient of friction of the wheat on the walls of the bin. As k increases, this ratio of height to diameter decreases. It would appear to be a serious objection to the use of [Equations \(5\)' and \(6\)'](#) if this ratio for maximum V was large, but it must be remembered that k , for earth over tunnels, is not known. It is possibly larger than assumed. In any case, [Equations \(9\)](#), which

were deduced independently of the modified Janssen formulas, appear to hold.

The writer has read with much interest the very interesting “dry sand and wheat arching experiments,” referred to by Mr. Meem. It is seen from the above, that the writer believes in this arching of sand under certain conditions, for example, after some settlement. He does not see any reason for any arching in an unlimited mass of sand, level at the top. The conjugate pressures here are vertical and horizontal; but, if a tunnel is bored through this mass, it tends to sink over the tunnel, and, only in consequence of that settlement, is a part of the weight of the sand directly over the tunnel transferred to the sides through the friction caused by the lateral thrust and the cohesion. Neither of these forces, both acting vertically upward, were in action, before the settlement. Mr. Meem gives the following [account](#) of an interesting experiment:

“A 2-in. pipe, 18 in. long, was filled with dry sand for a depth of 12 in., and a thin piece of tissue paper was pasted across the bottom. Then, with a wooden piston bearing on the sand, the latter would support the blow of a sledge hammer or the weight of a man without breaking the tissue paper.”

Considering the sand in the pipe alone, it affords a pretty illustration of the bin theory. Here, $R = \frac{A}{U} = \frac{\pi}{2\pi} = \frac{1}{2}$. Take $\phi = 30^\circ$ and $k\mu = \frac{10}{6} \tan^2 \left(45^\circ - \frac{\phi}{2} \right)$, $\tan \phi = 0.32$; also $w = \frac{90}{1728}$ lb. per cu. in. Therefore, making $c = 0$ in the formula for V above, we have $V = \frac{w R}{k \mu} = \frac{1}{0.32} \times \frac{45}{1728} = 0.0814$ lb. per sq. in. Hence the total pressure on the tissue paper is $\pi V = 0.256$ lb., or say $\frac{1}{4}$ lb. Perhaps the paper can stand this. The pressure is reduced to 0.185 lb. on the paper if we include cohesion, taking $c = 1$ lb. per sq. ft., as deduced from Leygue’s experiments on dry sand. This pressure would not be increased if the pipe, supposed to be vertical and filled with sand, was of great height, the weight

of the additional sand being equal to the weight of the man or to the pressure induced by the blow of the hammer. It seems natural, then, to infer that the pressures due to the blow or man, are sustained by the sides of the pipe, as in the case of the sand, though the conditions are not the same. In fact, in this case, the pressure on the paper is even less than before; for the blow, or the weight of the man causes the passive lateral thrust of the earth to be exerted, and this, for $\phi = 30^\circ$, is nine times the active thrust hitherto used, at least for an unlimited mass of earth. If this ratio is assumed to hold for the sand in the pipe, the value of $k \mu$ will be changed to 9×0.32 , and the total pressure on the paper will be only

$$\frac{1}{9} \times \frac{1}{4} = \frac{1}{36} \text{ lb.}$$

It is hoped that experimenters may turn their attention to finding definite values of the coefficient of cohesion for all kinds of earth. From observations of unsupported trenches, it has been seen that values of c of from 100 to possibly 400 lb. per sq. ft., may be expected. Résal states³² that MM. Jacquinot and Frontard, in July and August, 1910, made some preliminary experiments on earth taken from a reservoir dam which was failing, and found for it about $c = 2\,000$ kg. per sq. m., or say 409 lb. per sq. ft.; but $\tan \phi = 0.14$, corresponding to $\phi = 8^\circ$. The latter result is startling. For finding c and $\tan \phi$ experimentally, Résal suggests that a thin slice of earth be placed between two rough metallic plaques, pressed firmly together, and that the resistance to the relative displacement of the two plaques, for varying pressures, be recorded. By writing the relation between c , $\tan \phi$, and the forces involved, for each experiment, values of c and $\tan \phi$ can be found by elimination. In conclusion, the writer believes that he has offered a satisfactory and comprehensive theory of earth pressure, for earth endowed with both cohesion and friction. The results are not on as satisfactory a basis for pressures on tunnels, but the formulas derived are submitted in the hope that engineers will subject them to the test of both experience and experiment.

The writer returns sincere thanks to Messrs. Worcester and Meem for their helpful and stimulating discussion.

21 [Page 426.](#) [Return to text](#)

22 “The Bracing of Tunnels and Trenches, with Practical Formulas for Earth Pressures,” *Transactions*, Am. Soc. C. E., Vol. LX, p. 1; and “Pressure, Resistance, and Stability of Earth,” *Transactions*, Am. Soc. C. E., Vol. LXX, p. 352. [Return to text](#)

23 *Transactions*, Am. Soc. C. E., Vol. LX, p. 1. [Return to text](#)

24 *Transactions*, Am. Soc. C. E., Vol. LXX, p. 352. [Return to text](#)

25 See the writer’s “Retaining Walls,” sixth edition, p. 132. [Return to text](#)

26 *Transactions*, Am. Soc. C. E., Vol. LX, p. 84. [Return to text](#)

27 This is reviewed in *Engineering News*, January 19th, 1911. [Return to text](#)

28 The formulas used will be found in the writer’s “Retaining Walls,” sixth edition, p. 96, *et seq.* [Return to text](#)

29 This assumes that the values of k and c do not decrease for the greater depths. [Return to text](#)

30 In the Appendix, the “semi-empirical” formula, $L = (V - V') k$, was assumed. It gives fairly correct values for small values of y , but for c large and y large, it departs more from the truth than was at first surmised. Hence, by the reasoning above, V' will be made zero throughout, and [Equations \(5\)](#) and [\(6\)](#) of the Appendix will be changed to

$$V = \frac{1}{k\mu} [R w - c] \left(1 - e^{-\frac{k\mu}{R} y} \right) \quad (5)'$$

$$L = V k \quad (6)'$$

Equations (5)' and (6)' will not give accurate results for y small, which is of no importance, as such values are never used; but they should give practically accurate results for the larger values of y . [Return to text](#)

31 Ketchum’s “The Design of Walls, Bins and Grain Elevators,” pp. 253–282. [Return to text](#)

32 “*Poussée des Terres*,” *Deuxième Partie*, p. 327. [Return to text](#)

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